

flea

NEWS 88 *October 2021*



<http://esanetworks.org/>

FLEA NEWS is a twice-yearly newsletter about fleas (Siphonaptera). Before including in publications any of the citations given here, please review for accuracy and correct formatting. Many of our sources are abstracting journals and current literature sources, and citations have not necessarily been checked.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

R. L. Bossard, Ph.D.
Editor, Flea News
bossardtech@gmail.com

Organizers of the Flea News Network are Drs. R. L. Bossard and N. C. Hinkle.

N. C. Hinkle, Ph.D.
Dept. of Entomology
Univ. of Georgia
Athens GA 30602-2603 USA
NHinkle@uga.edu
(706) 583-8043

Assistant Editor J. R. Kucera, M.S.

Contents

Editorial, Send me flea species articles

Announcements

Featured research

Directory of Siphonapterists

Fleas in Art and History

Baroness Rosina Bulwer Lytton Lytton, *Cheveley: Or, The Man of Honour* (1839) (excerpt)

Michiel Sweerts,

A Sleeping Boy and a Man Removing Lice (or Fleas) from Himself (c. 1650 – 1654)

Siphonaptera Literature (2020) (continued), (2021)

Editorial

Send me flea species articles

Dear *Flea News* Reader,

Please send me articles, especially new flea species, and taxonomic revisions! Dr. Hastriter and I continue to update and correct the Lewis world flea species list.

The 2021 citations to date are in this *Flea News*, thanks to everyone sending me material.

Continued best wishes in regard to the coronavirus pandemic.

Yours in fleas,
R.L. Bossard
Editor, *Flea News*

**

Announcements

Royal Entomological Society Out of print handbook free download

HANDBOOKS FOR THE IDENTIFICATION OF BRITISH INSECTS

Vol 1 Part 16. Siphonaptera. F. G. A. M. Smit (1957)

<http://www.royensoc.co.uk/publications/index.htm>

<https://www.royensoc.co.uk/out-print-handbooks>

Smit's handbook on fleas is a well-written and illustrated, and helpful introduction to flea taxonomy, and it's now free for downloading; unfortunately, it deals only with British fleas (these are in 5 Families: Ceratophyllidae, Hystrihopsyllidae, Ischnopsyllidae, Leptopsyllidae, and Pulicidae), and it's out-of-date compared with Whitaker's (see below). Reprinting of a second edition of the Rothschild volumes appears unlikely in the near future even though there is demand. - Editor

"Out of stock RES Handbooks for the Identification of British Insects Volume 1 Part 16 (2nd edition) A.P.Whitaker, 2007 178pp"

"Since the first edition of this Handbook was published by F.G.A.M.Smit in 1957, six new species of Siphonaptera have been recorded in Britain, and one new family. An updated checklist of species found in the British Isles is given in this revised Handbook, which includes 62 species in 7 families [adds to Smit's CTENOPHTHALMIDAE, and VERMIPSYLLIDAE]. Keys to all families, genera and species, together with diagnostic illustrations, are provided to facilitate identification. 577 published distribution references are also listed, detailing the species of flea, and the host and county from which each was recorded."

**

Flea Presentations at the Entomological Society of America annual meeting 2020

Effects of mammalian community structure on ectoparasites in a desert ecosystem.

Jessica Taylor Martin, Richard E. Grewelle, IV, Ellen K. Bledsoe, Erica M. Christensen, Glenda M. Yenni, Gianalberto Losapio, Rodolfo Dirzo, and S.K. Morgan Ernest.

Fendona™ CS Controlled Release Insecticide efficacy evaluations for ticks and fleas on turf.

F. Carter Gordon, Jason Meyers, Robert W. Davis, Freder Medina, and Grzegorz Buczkowski.

Comparisons of ectoparasites of urban and wild mesomammals in north-central Florida.

Sarah Mays Maestas, Lindsay P. Campbell, Samantha M. Wisely, and Phillip E. Kaufman.

Rapid DNA detection in vector arthropods.

James Gerard Mann and Jason Pitts.

Investigating the effect of host characteristics on ectoparasite co-occurrence on two species of bats in Atlantic Canada.

Alexandra Sauk and Hugh G. Broders.

Fleas, blood-feeding and plague!

Viveka Vadyvaloo, Benjamin Burrows, and Kiara Held.

Prevalence of *Rickettsia* spp. and *Bartonella* spp. in cat fleas (*Ctenocephalides felis*) collected from feral/stray cats in Statesboro, Georgia.

Ryne Ward Maness and Lisa Diane Brown.

**

Featured Research 2020, 2021

New Flea Species 2020, 2021

Hastriter, M. W. (2021). Fleas (Siphonaptera: Pygiopsyllomorpha) of Papua New Guinea and Papua province (Indonesia). Part VI. *Bibikovana*, *Geohollandia*, and *Hoogstraalia* (Pygiopsyllidae: Pygiopsyllinae), with descriptions of four new species. *Annals of Carnegie Museum*, 87(1), 37-77. doi:10.2992/007.087.0102

Part VI is the final installment of a six-part series of papers discussing the fleas of Papua Province, Indonesia, and Papua New Guinea as a continued study of fleas in the Robert Traub flea collection deposited in the Carnegie Museum of Natural History, Pittsburgh, PA. Taxa from the genera *Bibikovana*, *Geohollandia*, and *Hoogstraalia* are included with emphasis on those species endemic to Papua Province, Indonesia, and Papua New Guinea. Four new species are described (*Bibikovana engilisi*, *Bibikovana bosaviensis*, *Bibikovana mekiliensis*, and *Hoogstraalia nadchatrami*). *Bibikovana laciniosa bismarckensis* is treated as a junior synonym of *Bibikovana laciniosa laciniosa* and *Hoogstraalia novaeguineae* is treated as a junior synonym of *Hoogstraalia turdella*. The females of *Geohollandia solida* and *Hoogstraalia turdella* are described and illustrated for the first time. The mammalian hosts of the three genera studied herein are summarized in Table 1. The avian host specificity of the genera *Geohollandia* and *Hoogstraalia* are substantiated with many new records of avian host associations included as Table 2. Geographic distributions of species of all three genera are summarized in Table 3. Keys for species of *Bibikovana* and *Hoogstraalia* and an updated key to the genera of Pygiopsyllomorpha are included. With the description of four new species and the proposed

synonymies of two species, the total number of species in the superfamily Pygiopsylloidea in Papua Province, Indonesia, Papua New Guinea (including Bismarck Archipelago) and the Solomon Islands is 113. An additional eight species belonging to three other flea families (Ischnopsyllidae (3), Pulicidae (3), and Leptopsyllidae (2) bring the total number of flea taxa to 121 species (including subspecies).

[*Bibikovana engilisi* Hastriter, 2021, *Bibikovana bosaviensis* Hastriter, 2021, *Bibikovana mekiliensis* Hastriter, 2021, and *Hoogstraalia nadchatrami* Hastriter, 2021]

Hastriter, M. W. (2021). Records of Fleas (Siphonaptera) from Australia, Malaysia, and Papua New Guinea with the Description of a New Species of *Bibikovana* Traub, 1980 (Pygiopsyllidae). *Annals of Carnegie Museum*, 87(2), 117-137.

The Robert Traub collection of fleas is maintained in the Carnegie Museum of Natural History. During the height of collecting fleas from the 1960s onward, many specimens were collected from Australia, Malaysia, and New Guinea and provided to Robert Traub. A plethora of publications resulted; however, Dr. Traub died before he could finish much of that work. Following his death, specimens were available that had never been fully studied, and in many cases, inadequately curated (alcohol material, unidentified slides, vials of dried up specimens, etc.). Following the in-depth reviews of 14 genera (published in six parts by the author), additional specimens were obtained from the Carnegie Museum. These additional specimens represented 32 species in nine flea families and are reported here for the first time. Many of the specimens included in this study, are supplementary specimens associated (but not reported) by Hastriter, and in Hastriter's six-part series (Parts I–VI) published between 2013 and 2021. Among the material was a new species of *Bibikovana*, from Australia, *Bibikovana acumena*, that is closely related to *Bibikovana colossus*. This new taxon is described and illustrated. The ranges of five flea species are extended: 1) *Porribius caminae* to Tasmania, 2) *Neopsylla dispar*, to the island of Borneo (State of Sabah, Malaysia), 3) *Bibikovana engilisi*, to Papua New Guinea, 4) *Pygiopsylla zethi* to Mole Creek and Bridport, Tasmania, and 5) *Smitella thambetosa*, to Papua New Guinea. Eight new host records include: 1) *Sigmactenus toxopeusi*, on *Rattus giluwensis* Hill, 1960, 2) *Stephanocircus pectinipes* Rothschild, 1915, on *Pseudomys novaehollandiae* (Waterhouse, 1843), 3) *Stephanocircus simsoni* Rothschild, 1905, on *Pseudomys higginsii* Trouessart, 1897, 4) *Bibikovana engilisi* Hastriter, 2021 on *Microperoryctes papuensis* (Laurie, 1952), 5) *Parastivalius gressitti* on *Mi. papuensis*, 6) *Rectidigitus glomerospinosus* Hastriter, 2016 on *Cercartetus caudatus* (Milne-Edwards, 1877), 7) *Rectidigitus traubi*, in a *Mi. papuensis* nest, and 8) *Smitella thambetosa* on *Mammelomys rattoides* (Thomas, 1922). Host data supporting the primary host of *Choristopsylla ochi* was presented and determined to be *Trichosurus vulpecula* (Kerr, 1792). Host associations are discussed for each species and a list of hosts and respective flea species are provided.

[*Bibikovana acumena* Hastriter, 2021]

Keskin, A. (2021). Description of a new flea species of the genus *Ctenophthalmus* Kolenati (Siphonaptera: Ctenophthalmidae) from Turkey. *Parasitology Research*, 120(4), 1203-1207.

A new ctenophthalmid flea species, *Ctenophthalmus (Ctenophthalmus) yesarii* n. sp., is described and illustrated. This new species belonging to the *fransmiti* group is most closely related to *Ctenophthalmus (Ctenophthalmus) fransmiti* Suciu. Specimens of *Ct. yesarii* n. sp. were collected from

the eastern broad-toothed field mouse, *Apodemus mystacinus* (Danford & Alston), in Balıkesir (Edremit district) province of Turkey. The number of *Ctenophthalmus* taxa known from Turkey now tallies 41 taxa (27 species and 14 subspecies).

[*Ctenophthalmus (Ctenophthalmus) yesarii* Keskin, 2021]

Keskin, A. (2020). A new flea species of the genus *Palaeopsylla* (Insecta: Siphonaptera: Ctenophthalmidae) from Turkey. *Journal of medical entomology*, 57(1), 88-91.

A new flea (Insecta: Siphonaptera) species belonging to the *Palaeopsylla* minor-group, *Palaeopsylla (Palaeopsylla) beaucournoi* n. sp. is described and illustrated. Specimens (14 males and seven females) of this new species were collected from *Talpa levantis* Thomas (Mammalia: Talpidae) in the deciduous mixed forest in Bursa province of Marmara Region, Turkey.

[*Palaeopsylla (Palaeopsylla) beaucournoi* Keskin, 2020]

Taxonomy

Meusemann, K., Trautwein, M., Friedrich, F., Beutel, R. G., Wiegmann, B. M., Donath, A., ... & Yeates, D. K. (2020). Are fleas highly modified Mecoptera? Phylogenomic resolution of Antliophora (Insecta: Holometabola). *BioRxiv*. (preprint not peer reviewed)

Insect orders have been defined and stable for decades, with few notable exceptions (e.g., Blattodea and Psocoptera). One of the few remaining questions of order-level monophyly is that of Mecoptera in respect to the phylogenetic placement of Siphonaptera (fleas). We used a large set of transcriptomic nucleotide sequence data representing 56 species and more than 3,000 single-copy genes to resolve the evolutionary history of Antliophora, including fleas (Siphonaptera), scorpionflies and relatives (Mecoptera), and true flies (Diptera). We find that fleas and mecopterans together are the sister group of flies. However, our data and/or analyses are unable to distinguish whether fleas are sister to a monophyletic Mecoptera, or whether they arose from within extant mecopteran families, rendering Mecoptera paraphyletic. We did not detect parameter bias in our dataset after applying a broad range of detection methods. Counter to a previous hypothesis that placed fleas within Mecoptera as the sister group to wingless boreids (snow fleas), we found a potential sister group relationship between fleas and the enigmatic family Nannochoristidae. Although we lack conclusive evidence, it seems possible that fleas represent the most-species rich group of modern mecopterans and that their parasitic lifestyle and morphological adaptations have simply made them unrecognizable in respect to their order-level classification.

Tihelka, Erik, M. Giacomelli, D. Huang, D. Pisani, P. Donoghue, and C. Cai. et al. "Fleas are parasitic

scorpionflies." *Palaeoentomology* 3.6 (2020): 641-653. DOI:
<https://doi.org/10.11646/palaeoentomology.3.6.16>

Fleas (Siphonaptera) are medically important blood-feeding insects responsible for spreading pathogens such as plague, murine typhus, and myxomatosis. The peculiar morphology of fleas resulting from their specialised ectoparasitic lifestyle has meant that the phylogenetic position of this diverse and medically important group has remained one of the most persistent problems in insect evolution. Here we test competing hypotheses on the contentious evolutionary relationships of fleas and antliophoran insects using the largest molecular dataset available to date consisting of over 1,400 protein-coding genes, and a smaller mitogenome and Sanger sequence alignment of 16 genes. By removing ambiguously aligned sequence regions and using site-heterogeneous models, we consistently recover fleas nested within scorpionflies (Mecoptera) as sister to the relictual southern hemisphere family Nannochoristidae. Topology tests accounting for compositional heterogeneity strongly favour the proposed topology over previous hypotheses of antliophoran relationships. This clade is diagnosed by shared morphological characters of the head and sperm pump. Fleas may no longer be regarded as a separate insect order and we propose that Siphonaptera should be treated as an infraorder within Mecoptera, reducing the number of extant holometabolan insect orders to ten.

[Editor's note: the two preceeding papers continue investigating the hypothesis of paraphyletic Mecoptera, with fleas descending from Mecopteran Boreidae (northern hemisphere snow scorpionflies) (Whiting 2002, *Zoologica Scripta* 31(1): 93–104, Whiting et al. 2008 A molecular phylogeny of fleas (Insecta : Siphonaptera): origins and host associations." *Cladistics* 24, no. 5. 677-707).

The phylogenetic placement of fleas stays up in the air with connections to the unusual relictual, southern hemisphere, living fossil Mecopteran Nannochoristidae (using amino acid transcriptome) contradicting amino acid and nucleotide results (Sanger sequences, mitogenomes) that link fleas to Boreidae (Whiting, et al. 2008) (Thihelka et al. specimens were *Boreus*, and *Caurinus dectes*), or to the aberrant, living fossil Mecopteran Eomeropidae (southern hemisphere) (specimen was *Notiothauma reedi*).

Stay tuned for further developments!]

**

KUMAR, RAHUL, SHASHANK, P.R., NARESH, M., MESHRAM, ANOOJ, S. S. & KOLLASREEDEVI. (2020) EVOLUTION OF MODERN INSECT CLASSIFICATION: A COMPREHENSIVE ACCOUNT. *J. Appl. Zool. Res.* 31(1): 1-15.

Insects are the most numerous and diverse group of animals in the entire animal kingdom. Insect classification has evolved in different phases over a considerable period of time. There are four different phases or eras, viz., the Pre-Linnean Era, the Linnean Era, the Darwinian Era, and the Hennigian Era and other developments. Among all eras the Hennigian Era marks a revolutionary shift in classification of insects with introduction of Phylogenetic Systematics that strictly considers

monophyly of different taxa to decide their respective positions within the tree of classification. Modern classification of insects is based on knowledge gained from multiple sources like Anatomy, Paleontology, Molecular Biology, and Embryology leading to a more natural and reliable classification scheme. Modern classification of insects also reflects their phylogeny. With new findings and advent of new technologies there is always a scope of change in the existing framework of insect classification in future.

**

Beaucournu J.-C., Ordre des Siphonaptères (Puces), Chapitre 37 in Aberlenc H., Les Insectes du Monde (2 volumes), Quae et Opie Ed., paru April 2021.

De Avelar, D.M.; Linardi, P.M. Siphonaptera. In: Andrade, H.T.A. & Serpa Filho, A. (org.). Princípios Básicos de Entomologia Médica. Natal, Brasil:Editora Caule de Papiro, 2021, p. 457-499.

Thoroughgood, J. T., Armstrong, J. S., White, B., Anstead, C. A., Galloway, T. D., Lindsay, L. R., . . . Chilton, N. B. (2021). Molecular differentiation of four species of *Oropsylla* (Siphonaptera: Ceratophyllidae) using PCR-based single strand conformation polymorphism analyses and DNA sequencing. *Journal of Medical Entomology*, 58(1), 241-245. doi:10.1093/jme/tjaa161

It is often difficult to distinguish morphologically between closely related species of fleas (Siphonaptera). Morphological identification of fleas often requires microscopic examination of internal structures in specimens cleared using caustic solutions. This process degrades DNA and/or inhibits DNA extraction from specimens, which limits molecular-based studies on individual fleas and their microbiomes. Our objective was to distinguish between *Oropsylla rupestris* (Jordan), *Oropsylla tuberculata* (Baker), *Oropsylla bruneri* (Baker), and *Oropsylla labis* (Jordan & Rothschild) (Ceratophyllidae) using PCR-based single strand conformation polymorphism (SSCP) analyses and DNA sequencing. A 446 bp region of the nuclear 28S ribosomal RNA (rRNA) gene was used as the genetic marker. The results obtained for 36 reference specimens (i.e., fleas that were morphologically identified to species) revealed no intraspecific variation in DNA sequence, whereas the DNA sequences of the four species of *Oropsylla* differed from one another at two to six nucleotide positions. Each flea species also had a unique SSCP banding pattern. SSCP analyses were then used to identify another 84 fleas that had not been identified morphologically. DNA sequencing data confirmed the species identity of fleas subjected to SSCP. This demonstrates that PCR-SSCP combined with DNA sequencing of the 28S rRNA gene is a very effective approach for the delineation of four closely related species of flea.

Zurita, A., & Cutillas, C. (2021). Combination of nuclear and mitochondrial markers as a useful tool to identify *Ctenophthalmus* species and subspecies (Siphonaptera: Ctenophthalmidae). *Organisms Diversity and Evolution*, doi:10.1007/s13127-021-00500-4

Ctenophthalmus is considered the largest genus within the Order Siphonaptera. From a morphological point of view, only males of this genus can be identified at species and subspecies levels using morphological keys, whereas there are no morphological criteria in order to classify females at these taxonomical levels. Furthermore, the amount of available molecular and phylogenetic data for this genus is quite scarce so far. The main objective of this work was to assess the utility of the combination of nuclear and mitochondrial markers with respect to their ability to differentiate among different subspecies within the *Ctenophthalmus* genus. With this purpose, we carried out a comparative morphological and molecular study of three different subspecies (*Ctenophthalmus baeticus arvernus*, *Ctenophthalmus nobilis dobyi*, and *Ctenophthalmus andorrensis catalaniensis*) in order to clarify and discuss its taxonomic status. In addition, our study complemented the molecular data previously provided for *Ctenophthalmus baeticus boisseauorum* and *Ctenophthalmus apertus allani* subspecies. We sequenced five different molecular markers: EF1- α , ITS1, ITS2, cox1, and cytb. Our results confirmed that morphological data by themselves are not able to discriminate among *Ctenophthalmus* female taxa; however, the combination of the nuclear marker EF1- α together with mtDNA markers cytb and cox1 constituted a useful taxonomical and phylogenetic tool to solve this issue. Based on these results, we consider that the use of this molecular approach should be gradually used within *Ctenophthalmus* genus in order to complement its classical taxonomy and clarifying the complex taxonomy of other congeneric species of fleas.

Keywords Siphonaptera . *Ctenophthalmus* . Taxonomy . Mitochondrial DNA . Nuclear DNA

Berenbaum, M. R. (2021). Flea speech. American Entomologist, 67(2), 10-13. doi:10.1093/ae/tmab030 [Rothschild quote]

Bernard, E. C., & Whittington, A. E. (2021). Papers and new species of minor insect orders published in *Zootaxa*, 2001–2020. *Zootaxa*, 4979(1), 232-235. doi:10.11646/zootaxa.4979.1.26

A summary is presented of papers published on minor insect orders (MIO) in *Zootaxa*'s first 20 years, as well as the number of new species described therein. The MIO orders currently covered by the editors and summarized here include **Archaeognatha** (Microcoryphia), **Dermaptera**, **Embioptera**, **Siphonaptera**, **Zoraptera** and **Zygentoma**, as well as the hexapod classes **Protura** and **Diplura**. Both fossil and extant taxa of these groups are included in the MIO purview. The MIO editors also have frequently served as interim editors for groups temporarily without a subject-matter editor, such as Carabidae, Blattodea and Mantodea; as a backup editor for Mecoptera; and as ad-hoc editors for papers written by the editors of a different taxonomic group or for papers without a single-taxon focus. In the period 2001–2020, descriptions of 130 new species were published, compared with 816 species in all other journals. The greatest number of species were for Protura (49), while Zoraptera had the highest proportion of papers compared to all publications (50%).

[Editor's note: the authors show an average of 10 new flea species described each year from 2001 to 2020, with 10+ flea experts worldwide.]

Zhang, Y., Nie, Y., Deng, Y. -, Liu, G. -, & Fu, Y. -. (2021). The complete mitochondrial genome sequences of the cat flea *Ctenocephalides felis felis* (Siphonaptera: Pulicidae) support the hypothesis that *C. felis* isolates from China and USA were the same *C. f. felis* subspecies. *Acta Tropica*, 217 doi:10.1016/j.actatropica.2021.105880

Hosts and Distribution

Beard, D., Stannard, H. J., & Old, J. M. (2021). Parasites of wombats (family Vombatidae), with a focus on ticks and tick-borne pathogens. *Parasitology Research*, 120(2), 395-409. doi:10.1007/s00436-020-07036-0

Berrizbeitia, M. F. L., Hastriter, M. W., Sanchez, J. P., & Díaz, M. M. (2021). New record of *Alectopsylla unisetosa* (Siphonaptera: Ischnopsyllidae) from Patagonia. *Parasitology International*, 102426.

Physiology

Bossard, R. L. 2021. Possible thermal and respiratory constraints for fleas (Siphonaptera: Ceratophyllidae) on North American deer mice (Mammalia: Rodentia). *Journal of the Utah Academy of Science, Arts, and Letters* (in press) (abstract). (not peer reviewed)

There are numerous examples of terrestrialization and diversification of ceratophyllid fleas in North America. The flea genus *Aetheca* is confined to western North America and its 2 species appear currently nearly allopatric. *Aetheca thamba* is at high elevations in the U.S. Rocky Mountains, and lower elevations in Canada, but *Aetheca wagneri* is at lower elevations in the western U.S. Parapatric thermal speciation may have occurred since the last Ice Age, because deer-mice hosts (*Peromyscus* spp.) are abundant throughout these fleas' ranges. Thermal speciation and thermal niches are reported in the ocean and the terrestrial tropics for diverse phyla. In addition to direct effects of temperature on the growth of larval fleas, correlated processes such as respiration may constrain flea occurrence. The flea *Orchopeas leucopus* co-occurs with *A. wagneri* in the Great Basin Desert, often infesting the same individual mouse. However, *O. leucopus* appears constrained to cool, moist times of the year, but with a widespread distribution including eastern North America, unlike *A. wagneri*, which occurs throughout the year but only in western North America. The contrasting geography and seasonality of *O. leucopus* and *A. wagneri* may be partly a result of thermal and respiratory constraints peculiar to their respective phylogenies. The importance of competition in mediating abiotic factors among flea larvae or among flea adults is unclear. *Aetheca* and *Orchopeas* are both Ceratophyllidae, a new flea family possibly originating in the Himalaya Mountains, which radiated explosively all through the Holarctic region, by colonizing new hosts in extreme environments by using unknown adaptations [however, see Editor's

note below]. Flea respiratory systems are exceedingly diverse, dynamic, and phylogenetically conserved, but have not been linked clearly to species habitat or seasonality. Fleas on deer-mice are an exciting community for understanding physiological ecology.

[Editor's note: the location where Ceratophyllidae originated is unknown, but could be Nearctic. Traub, et al. (The Rothschild Collection of Fleas: The Ceratophyllidae, pg. 186) wrote: "The available data on the host-relationships and zoogeography of the Ceratophyllidae strongly suggest that the Sciurids are the aboriginal hosts of this group of fleas and both arose in the Nearctic Region". Medvedev (1996) noted: "The family Ceratophyllidae has thus a distinct Asian-American distribution, resulting from the existence of the Beringian bridge. It is difficult to locate the place of origin of this family without additional morphological analysis", and that most species of this family are Palearctic. Closely related families in the infraorder Ceratophyllomorpha containing Ceratophyllidae likely originate in the Palearctic.

Krasnov et al. (2015) found evidence of multiple flea colonizations in both directions (east to west, west to east) between Nearctica and Palearctica (Krasnov, B. R., Shenbrot, G. I., & Khokhlova, I. S. (2015). Historical biogeography of fleas: the former Bering Land Bridge and phylogenetic dissimilarity between the Nearctic and Palearctic assemblages. *Parasitology research*, 114(5), 1677-1686.). The Ceratophyllidae are "the most recently evolved family of Siphonaptera, probably dating to very early Oligocene, perhaps 35 to 40 million years ago" (Traub et al. pg. 188).]

**

Bossard, R. L. 2021. Seasonality of 4 prevalent flea species (Siphonaptera: Ceratophyllidae, Leptopsyllidae) on North American deer-mice *Peromyscus maniculatus* (Mammalia: Rodentia) due to possible thermal-niche partitioning. Ecological Society of America annual meeting, August 2-6, 2021, Long Beach, California, USA 2021 (poster). DOI: 10.13140/RG.2.2.28758.37448 (not peer reviewed)

Temperature was described as an axis of the niche in 1979, and in the decades since, thermal-niches have been discovered in many diverse taxa of bacteria, protists, fungi, plants, and animals including endothermic vertebrates. Do fleas show evidence of thermal niches? My objective was to analyze seasonal prevalence of adult fleas on deer-mice in relation to ambient temperature, and to determine if the occurrences of flea species were interrelated. Prevalence of the common flea species on *Peromyscus maniculatus* was tabulated from trapping in the Great Basin Desert by H.J. Egoscue, with optimal prevalence estimated by fitting LRJ [sic, LRF] non-linear curves versus estimated monthly temperatures near the trap locations.

The 4 most prevalent flea species on *P. maniculatus* - *Aetheca wagneri*, *Opisodasys keeni keeni* [sic, *Opisodasys keeni*], *Orchopeas leucopus*, and *Peromyscopsylla hesperomys adelpha* - display characteristic temperature optima and ranges minimizing their overlap on the temperature axis. In conclusion, the common flea species on North American deer-mice in the Great Basin Desert show distinct thermal niches, and because seasonal separation of flea species shows up in many flea hosts, including flying squirrels, gerbils, and ground squirrels, thermal-niche partitioning may be a widespread device that sustains a diverse flea community on a host under varying temperatures.

**

Bossard, R.L. 2021. Temperature Phenology of Fleas (Siphonaptera) on Rodents (Mammalia: Rodentia). Entomological Society of America annual meeting, October 31-November 3, 2021, Denver, Colorado USA (poster). (not peer reviewed)

Seasonality of fleas (Siphonaptera) is often attributed to competition, prompting the idea that flea species compete for, and partition, temperature as a resource, along with correlated variables such as moisture, involving phylogenetic conservatism, character displacement, and phenotypic plasticity (Thermal-niche Partitioning Hypothesis). Do the common flea species in a community on a host show distinct, partitioned thermal niches? I compared 5 rodent-flea communities for thermal niche optima by fitting non-linear LRF (Lobry–Rosso–Flandrois) curves, and examined ecoregion and climate for each community. Hosts were Southern flying squirrel (*Glaucomys volans volans*), Richardson's ground-squirrel (*Spermophilus richardsonii*), North American deer-mouse (*Peromyscus maniculatus*), Midday jird (*Meriones meridianus*), and Wagner's gerbil (*Dipodillus [=Gerbillus] dasyurus*).

Results for mesic flea communities showed distinctly spaced thermal-niche optima, supporting the Thermal-niche Partitioning Hypothesis; however, the xeric flea communities did not. In xeric communities, rodent fleas may compete for resources other than temperature, such as moisture or host availability, and for jird fleas, varied thermal-niche widths, not optima, create flea seasonality. Variations in seasonal timing among xeric fleas may correspond with desiccation resistance, while mesic fleas appear to compete for temperature, resulting in thermal-niche partitioning and seasonality, because resources such as moisture are less limiting. Non-competition seasonality hypotheses (Environmental Filtering, Neutrality) require testing, too. Thermal niche partitioning may increase "species-packing" of disparate fleas from many families onto a host, increasing the host's flea-species richness. Thermal niches may partition many insect and plant communities, so implications for biodiversity conservation and disease ecology changing under global warming are wide-ranging.

**

Veitch, J. S. M., Bowman, J., Mastromonaco, G., & Schulte-Hostedde, A. I. (2021). Corticosterone response by *Peromyscus* mice to parasites, reproductive season, and age. *General and Comparative Endocrinology*, 300 doi:10.1016/j.ygcen.2020.113640

Evolution

Driscoll, T. P., Verhoeve, V. I., Gillespie, J. J., Johnston, J. S., Guillotte, M. L., Rennoll-Bankert, K. E., . . . Azad, A. F. (2020). A chromosome-level assembly of the cat flea genome uncovers rampant gene duplication and genome size plasticity. *BMC Biology*, 18(1) doi:10.1186/s12915-020-00802-7

Background: Fleas (Insecta: Siphonaptera) are small flightless parasites of birds and mammals; their blood-feeding can transmit many serious pathogens (i.e., the etiological agents of bubonic plague, endemic and murine typhus). The lack of flea genome assemblies has hindered research, especially comparisons to other disease vectors. Accordingly, we sequenced the genome of the cat flea,

Ctenocephalides felis, an insect with substantial human health and veterinary importance across the globe.

Results: By combining Illumina and PacBio sequencing of DNA derived from multiple inbred female fleas with Hi-C scaffolding techniques, we generated a chromosome-level genome assembly for *C. felis*. Unexpectedly, our assembly revealed extensive gene duplication across the entire genome, exemplified by ~38% of protein-coding genes with two or more copies and over 4000 tRNA genes. A broad range of genome size determinations (433–551 Mb) for individual fleas sampled across different populations supports the widespread presence of fluctuating copy number variation (CNV) in *C. felis*. Similarly, broad genome sizes were also calculated for individuals of *Xenopsylla cheopis* (Oriental rat flea), indicating that this remarkable “genome-in-flux” phenomenon could be a siphonapteran-wide trait. Finally, from the *C. felis* sequence reads, we also generated closed genomes for two novel strains of *Wolbachia*, one parasitic and one symbiotic, found to co-infect individual fleas.

Conclusion: Rampant CNV in *C. felis* has dire implications for gene-targeting pest control measures and stands to complicate standard normalization procedures utilized in comparative transcriptomics analysis. Coupled with co-infection by novel *Wolbachia* endosymbionts—potential tools for blocking pathogen transmission—these oddities highlight a unique and underappreciated disease vector.

**

Azrizal-Wahid, N., Sofian-Azirun, M., & Low, V. L. (2020). New insights into the haplotype diversity of the cosmopolitan cat flea *Ctenocephalides felis* (Siphonaptera: Pulicidae). *Veterinary parasitology*, 281, 109102.

Highlights

Moderate genetic differentiation and high gene flow were observed in cat flea populations in Malaysia.

Three aberrant haplotypes in *cox2* sequences were observed, suggesting introgression or presence of cryptic species.

Two major lineages by which Malaysian haplotypes were closely related with haplotypes from the tropical region.

The present study investigated the genetic profile of the cosmopolitan cat flea, *Ctenocephalides felis* (Siphonaptera: Pulicidae) from Malaysia and the reference data available in the National Center for Biotechnology Information (NCBI) GenBank. A set of sequences of 100 Malaysian samples aligned as 550 characters of the cytochrome c oxidase subunit I (*cox1*) and 706 characters of the II (*cox2*) genes revealed ten haplotypes (A1–A10) and eight haplotypes (B1–B8), respectively. The concatenated sequences of *cox1* and *cox2* genes with a total of 1256 characters revealed 15 haplotypes (AB1–AB15). Analyses indicated that haplotype AB1 was the most frequent and the most widespread haplotype in Malaysia. Overall haplotype and nucleotide diversities of the concatenated sequences were 0.52909 and 0.00424, respectively, with moderate genetic differentiation ($F_{ST} = 0.17522$) and high gene flow ($N_m = 1.18$). The western population presented the highest genetic diversity ($H_d = 0.78333$, $P_i = 0.01269$, $N_h = 9$), whereas the southern population demonstrated the lowest diversity

($H_d = 0.15667$, $P_i = 0.00019$, $N_h = 3$). The concatenated sequences showed genetic distances ranged from 0.08 % to 4.39 %. There were three aberrant haplotypes in *cox2* sequences that highly divergent, suggesting the presence of cryptic species or occurrence of introgression. In the global point of view, the aligned sequences of *C. felis* revealed 65 haplotypes (AA1–AA65) by the *cox1* gene ($n = 586$), and 27 haplotypes (BB1–BB27) by the *cox2* gene ($n = 204$). Mapping of the haplotype network showed that Malaysian *C. felis* possesses seven unique haplotypes in both genes with the common haplotypes demonstrated genetic affinity with *C. felis* from Southeast Asia for *cox1* and South America for *cox2*. The topologies of *cox1* and *cox2* phylogenetic trees were concordant with relevant grouping pattern of haplotypes in the network but revealed two major lineages by which Malaysian haplotypes were closely related with haplotypes from the tropical region.

**

Moura, J. F., Nascimento, C. S. I., Peixoto, B. D. C. P. E. M., de Barros, G. E. B., Robbi, B., & Fernandes, M. A. (2021). Damaged armour: Ichnotaxonomy and paleoparasitology of bioerosion lesions in osteoderms of Quaternary extinct armadillos. *Journal of South American Earth Sciences*, 109 doi:10.1016/j.jsames.2021.103255

[Editor's note: Ichnotaxonomy attempts to group living things by their traces, tracks, holes, and burrows; the "fossilized results of behavior" (Magwood, J.P.A. 2017. *Short courses in Paleontology* 5:15-33.0].

**

Ohnishi, O., Takenaka, M., Okano, R., Yoshitomi, H., & Tojo, K. (2021). Wide-scale gene flow, even in insects that have lost their flight ability: Presence of dispersion due to a unique parasitic ecological strategy of piggybacking hosts. *Zoological Science*, 38(2), 122-139. doi:10.2108/zs200088

Ecology

Medvedev, S. G., Sedikhin, N. V., & Krasnov, B. R. (2020). Intraspecific variation of body size in fleas: Effects of host sex and flea phenology. *Parasitology Research*, 119(10), 3211-3220. doi:10.1007/s00436-020-06867-1

Krasnov, B. M., Stanko, M., Lareschi, I. S., Khokhlova. 2020. Species co-occurrences in ectoparasite infracommunities: accounting for confounding factors associated with space, time, and host community composition. *Ecological Entomology* 45: 1158-1171. DOI:10.1111/een.12900

Gozzi A.C., Marcela Lareschi, G. Navone, M.L. Guichón. 2020. The Enemy Release Hypothesis and *Callosiurus erythraeus* in Argentina: combining community and biogeographical parasitological studies. *Biological Invasions* 22 (12). DOI 10.1007/s10530-020-02339-w

Urdapilleta M., Galliari, C.A., Navarro Febre, T, & M. Lareschi. 2021. Effect of host and environment related factors on the distribution of the ectoparasites of the montane grass mouse *Akodon montensis* (Cricetidae, Sigmodontinae) in the Atlantic Forest ecoregion in northeastern Argentina. *Revista Mexicana de Biodiversidad*. En prensa.

Pathology and Control

Azrizal-Wahid, N., Sofian-Azirun, M., & Low, V. L. (2021). Flea-borne pathogens in the cat flea *Ctenocephalides felis* and their association with mtDNA diversity of the flea host. *Comparative Immunology, Microbiology and Infectious Diseases*, 75 doi:10.1016/j.cimid.2021.101621

Wong, Serena, Hany Elsheikha, and Michael Dryden. 2021. Flea product efficacy, pet owners' adherence and treatment failure: what's the connection? *Companion Animal* 26:8, 182-190.

Harvey T.V., Linardi P.M., Carlos R.S.A. & Heukelbach J. 2021. Tungiasis in domestic, wild, and synanthropic animals in Brazil. *Acta Tropica*. doi.org/10.1016/j.actatropica. 2021.106068.

Medvedev, S. G., Verzhutsky, D. B., & Kotti, B. K. (2020). Diversity of vectors of plague pathogens: Polyhostal parasites, fleas of the genus *Rhadinopsylla* Jordan et Rothschild, 1911 (Siphonaptera, Hystrihopsyllidae). *Entomological Review*, 100(9), 1218-1235. doi:10.1134/S001387382009002X

Medvedev, S. G., & Verzhutsky, D. B. (2020). Diversity of fleas, vectors of plague pathogens: The flea *Oropsylla silantiewi* (Wagner, 1898) (Siphonaptera, Ceratophyllidae). *Entomological Review*, 100(1), 45-57. doi:10.1134/S0013873820010054

The taxonomic diversity, distribution patterns, and host associations of fleas of the genus *Oropsylla*, which are vectors of the plague pathogen in the Eurasian and North American plague foci, are analyzed. Data on the abundance, ecological features, and epizootological significance of the marmot parasite *Oropsylla silantiewi* (Wagner, 1898) in different Palaearctic plague foci are summarized.

Cooke, B. D. (2021). Pasture plant biomass increase following introduction of European rabbit fleas, *Spilopsyllus cuniculi*, into Australia to facilitate myxomatosis transmission. *Biological Control*, 155 doi:10.1016/j.biocontrol.2021.104536

Kerr, P. J., Hall, R. N., & Strive, T. (2021). Viruses for landscape-scale therapy: Biological control of rabbits in Australia doi:10.1007/978-1-0716-1012-1_1

Microbial Symbiotes

- Urdapilleta M., Cicuttin GL, De Salvo MN, Pech May A, Salomon OD & Lareschi, M. 2020. Molecular detection and identification of *Bartonella* in the cat flea *Ctenocephalides felis felis* collected from cats and dogs in a border area in northeastern Argentina. Veterinary Parasitology: Regional Studies and Reports 19: 100361 <https://doi.org/10.1016/j.vprsr.2019.100361>.
- Moreno Salas L., M. Espinoza-Carniglia, N. Lizama-Schmeisser, L. G. Torres Fuentes, C. Silva de La Fuente, M. Lareschi and D. González-Acuña. 2020. Molecular detection of *Rickettsia* in fleas from micromammals in Chile. Parasite & Vectores, 13: 523. DOI: 10.21203/rs.3.rs-35524/v1
- Melis M., M. Espinoza-Carniglia, E. Savchenko, S. Nava, M. Lareschi. 2020. Molecular detection and identification of *Rickettsia felis* in a *Polygenis* flea (Siphonaptera, Rhopalopsyllidae, Rhopalopsyllinae) associated with cricetid rodents from a rural area in central Argentina. Veterinary Parasitology: Regional Studies and Reports 21 (2020): 100445
- Zurita, A., Benkacimi, L., El Karkouri, K., Cutillas, C., Parola, P., & Laroche, M. (2021). New records of bacteria in different species of fleas from France and Spain. Comparative Immunology, Microbiology and Infectious Diseases, 76 doi:10.1016/j.cimid.2021.101648

**

Directory of Siphonapterists (updated)

Dra. Roxana Acosta-Gutierrez
 Museo de Zoología
 Departamento de Biología Evolutiva
 Facultad de Ciencias, UNAM
 Apo. Postal 70-399,
 C.P. 04510, Mexico. D.F.
 roxana_a2003@yahoo.com.mx
 Flea biogeography, systematics, and taxonomy.

Dr. J.C. Beaucournu
 Parasitologie medicale
 Faculte de Medecine de Rennes
 2, avenue du Professeur Leon Bernard
 F 35043 Rennes cedex
 France
jc.beaucournu@gmail.com

Dr. Marian Blaski
 Silesian University, Department of Zoology
 ul. Bankowa 9
 40-007 Katowice
 Poland
marian.blaski@us.edu.pl

Dr. R.L. Bossard
 Bossard Consulting
 Salt Lake City, UT
bossardtech@gmail.com
 Ecology of host-parasite relationships.

Dra. Cristina Cutillas
 Departamento de Microbiología y Parasitología
 Facultad de Farmacia
 Universidad de Sevilla
 C/ PROFESOR GARCÍA GONZÁLEZ, Nº 2
 41012 Sevilla. Spain
cutillas@us.es
 Molecular biology of fleas

Dr. Anne Darries-Vallier
 Bio Espace – Laboratoire d'Entomologie
 Mas des 4 Pilas - Route de Bel-Air
 34570 Murviel les Montpellier
 France
bioespace.labo@gmail.com
 Ecology, biology, behavior, mass rearing, disease vectors.

Bill Donahue, PhD
 Sierra Research Laboratories
 5100 Parker Road
 Modesto, CA 95357
www.sierraresearchlaboratories.com
bill@sierraresearchlaboratories.com

Michael W Dryden, DVM, PhD, DACVM
 University Distinguished Professor of Veterinary Parasitology
 Department of Diagnostic Medicine/Pathobiology
 202 Trotter Hall
 Manhattan, KS 66506
 Dryden@vet.k-state.edu
 Veterinary parasitology.

Thomas M. Dykstra, Ph.D.
 Dykstra Laboratories, Inc.
 3499 NW 97th Blvd., Suite 6
 Gainesville, FL 32606
www.dykstralabs.com
 dykstralabs@yahoo.com
 Sensory perception of fleas, especially larvae.

Lance A. Durden, Ph.D.
 Department of Biology
 Georgia Southern University
 69 Georgia Avenue
 P. O. Box 8042
 Statesboro, Georgia 30460, USA
 ldurden@georgiasouthern.edu
 Flea taxonomy and host associations, especially in eastern North America and the Indo-Australian region.

Laura Fielden, Ph.D.
 Associate Professor, Department of Biology
 Truman State University, Kirksville, MO 63501
 lfielden@truman.edu
 Host specificity of fleas.

Manuel Fabio Flechoso del Cueto
 C/ Heroes de la Independencia no 1 - 2oA
 42200 Almazan (Soria), SPAIN
 fabioflechoso@hotmail.com

Dr. Patrick Foley
 Department of Biological Sciences
 California State University
 Sacramento, CA 95819
 patfoley@csus.edu
 Epidemiology, extinction, metapopulations, identification of fleas, plague.

Dr. Terry Galloway
 Professor of Entomology and Associate Curator
 J.B. Wallis Museum of Department of Entomology
 Winnipeg, Manitoba
 Canada R3T 2N2
Terry_Galloway@umanitoba.ca
 Ecology and taxonomy, especially of the larvae.

Dr. N. C. Hinkle
 Professor
 Dept. of Entomology
 Univ. of Georgia
 Athens, GA 30602-2603
NHinkle@uga.edu
 Flea biology and control.

Simon Horsnall
 UK Flea Recorder
simon.horsnall@googlemail.com

Adem Keskin, Ph.D.
 Assoc. Prof.
 Department of Biology, Faculty of Science and Art,
 Tokat Gaziosmanpaşa University, 60250, Tokat, Turkey
ademkeskin@yahoo.com
 Taxonomy, host associations, and vectorial importance of fleas.

Kerv Hyland
 5 Timber Lane, Unit 314
 Exeter, NH 03833
khyland@etal.uri.edu
 All groups of ectoparasites on vertebrates.

James (Jim) R. Kucera, M.S.
 5930 Sultan Circle
 Murray, UT 84107-6930
jameskucera@aol.com
 Flea systematics & taxonomy, host relationships, distribution & biogeography.

Dra. Marcela Lareschi
 Centro de Estudios Parasit. Vectores
 CEPAVE (CONICET-UNLP)
 Calle 2 # 584
 La Plata (1900)
 Argentina
mlareschi@cepave.edu.ar

Dr. Pedro Marcos Linardi
 Professor of Parasitology and Medical Entomology
 Departamento de Parasitologia
 Instituto de Ciencias Biologicas
 Universidade Federal de Minas Gerais
 31.270-901 Belo Horizonte, Minas Gerais
 Brasil
 linardi@icb.ufmg.br
 Taxonomy of fleas, host-parasite relationships, fleas as vectors of parasites.

Dr. Erica McAlister
 Department of Life Sciences
 The Natural History Museum
 Cromwell Road LONDON SW7 5BD UK
 Fleas at the NHM.

Christine M. McCoy, M.S.
 Staff Entomologist
 BerTek, Inc.
 Greenbrier, AR 72058
 cmccoy@bertekconsults.com

Sergei G. Medvedev, Doctor of Biology
 Chief of the Department of Parasitology
 Zoological Institute of Russian Academy of Sciences
 Universitetskaya Embankment 1
 St. Petersburg, 199034 Russia
<http://www.zin.ru/Animalia/Siphonaptera/index.htm>
 fleas@zin.ru
 Flea taxonomy and phylogenetics.

Dr. Norbert Mencke
 Grunder Muehle 2 B
 51381 Leverkusen
 Germany
 mencke_norbert@yahoo.com
 Veterinary specialist for parasitology.

Ettore Napoli
 Department of Veterinary Science unit Parasitology
 University of Messina
 Messina, Italy
 enapoli@unime.it
 Ectoparasitic arthropods and their vectorial role.
 MDV PhD student - Public Health

Dr. Barry M. Oconnor, Curator & Professor
 Museum of Zoology, University of Michigan
 1109 Geddes Ave
 Ann Arbor, MI 48109-1079
bmoc@umich.edu
 Ectoparasites of vertebrates, particularly mites.

Dra. Juliana P. Sanchez
 Centro de Investigaciones y Transferencia del Noroeste de la Provincia de Buenos Aires (CITNOBA-CONICET)
 Ruta Provincial 32 Km 3.5
 2700 Pergamino
 Buenos Aires, Argentina.
julianasanchez@unnoba.edu.ar
 Ectoparasite (especially flea) systematics, taxonomy, and ecology.

Jeff Shryock, M.S.
 Sr. Research Biologist
 Merial, Ltd., Missouri Research Center
 6498 Jade Rd.
 Fulton, MO 65251
Jeffrey.shryock@merial.com

Dr. Andrew Smith
 Vet and Biomedical Sciences
 Murdoch University
 Murdoch 6150
 West Australia
Andrew.Smith@murdoch.edu.au
 Ecology; fleas as vectors of parasites and associated diseases.

Jasmine Veitch, MSc
 Research Intern - Centre for Evolutionary Ecology and Ethical Conservation (CEEEC) (www.ceeec.ca)
 Department of Biology, Laurentian University 935 Ramsey Lake Rd., Sudbury, ON P3E2C6, Canada
jveitch1@laurentian.ca
 Evolutionary ecology and community ecology.

Dr. Amoret P. Whitaker, BSc MSc DIC DipFMS
 Scientific Associate – Forensic Entomology
 Department of Life Sciences
 Natural History Museum
 Cromwell Road London SW5 7BD
a.whitaker@nhm.ac.uk
 Forensic entomology.

Bill Wills
 Adj. Professor
 142 Heritage Village Lane
 Columbia, South Carolina 29212
 pb.wills@att.net
 Flea ecology.

**

Fleas in Art and History

Cheveley: Or, The Man of Honour

by Baroness Rosina Bulwer Lytton Lytton

(b. 4 November 1802, Ballywhire, County Limerick, Ireland,
 d. 12 March 1882, Upper Sydenham, London, England)

Volume 1 (excerpt), Baudry's European Library, Paris.
 Harper & Bros., 1839

[Editor's note: sources of this excerpt have different punctuations, so I combined them.]

“Well,” said Rush, drawing the precious *morceau* from his waistcoat pocket, where it had lain like a love-letter in a post-office, 'to be kept till called for,' – “remember it's not *finished*, and it's an irregular ode,”—and irregular enough it certainly was,—“and, in order to make you understand it, I should tell you that one flea enacts Napoleon, another the Duke of Wellington, two more O'Connell and Lord Alvanley fighting a duel, a whole band imitate Strauss's, while two young fleas are flirting in a drawing-room, and the mamma flea is reading a newspaper and does not see them.” After this clear and lucid explanation, Mr. Rush smoothed the crumpled ode, and read as follows, his mother exclaiming, by way of prelude, “How very interething !”

ODE TO THE INDUSTRIOUS FLEAS!

Silent ! subtle ! hopping flea !
 Napoleon's self may be in thee ;
 For still thou dost présent to me
 A marvel and a mystery !
 But Wellington thou canst not be,
 Since 'tis well known *he* ne'er did *flee*.
 Then jump and bite,
 And take thy flight,
 Like Joseph, when from Zulica, he
 Did take his garments up and flee,
 While sages say and shake their head,
 “ 'T was Fleance did it, for 't was Fleance fled.”
 If blame you'd shun,

Ne'er cut and run.
 But what is this? Oh fire and fuel !
 Next see O'Connell in a duel,—
 Nay, surely, sir, the mighty Dan
 May kill, but never flee his man.
 Oh keep, St. Kevin,
 His vow in heaven !
 But list! Flea fiddlers fill the house,
 With sounds not quite so good as Strauss';
 A damsel's in a ball-room flirting,
 Her mother's chap'ronage deserting;
 A flea in ear
 She'll get, I fear!

"That's all I've done yet, " said Mr. Rush, looking round for admiration.

"Oh, very clever—very clever indeed," cried Mrs. Tymmons ; " in fact, tho clever, that ith not every one that would underthand it."

" It's a great deal cleverer than that," said Hoskins, " for augh don't think there's ony one who could understand it."

**

On the next page

A Sleeping Boy and a Man Removing Lice (or Fleas) from Himself (c. 1650 – 1654)

by Michiel Sweerts

(Flemish, b. 29 September 1618, Brussels, now Belgium, d. 1 June 1664, Goa, India)

Oil painting on canvas (78.5 × 71 cm)(30.9 × 27.9 in)



Siphonaptera Literature (2020) (continued)

KUMAR, RAHUL, SHASHANK, P.R., NARESH, M., MESHRAM, ANOOJ, S. S. & KOLLASREEDEVI. (2020) EVOLUTION OF MODERN INSECT CLASSIFICATION: A COMPREHENSIVE ACCOUNT. J. Appl. Zool. Res. 31(1): 1-15.

Meusemann, K., Trautwein, M., Friedrich, F., Beutel, R. G., Wiegmann, B. M., Donath, A., ... & Yeates, D. K. (2020). Are fleas highly modified Mecoptera? Phylogenomic resolution of Antliophora (Insecta: Holometabola). BioRxiv. (preprint not peer reviewed)

Tihelka, Erik, M. Giacomelli, D. Huang, D. Pisani, P. Donoghue, C. Cai. et al. Fleas are parasitic scorpionflies. Palaeoentomology 3.6 (2020): 641-653.

<https://doi.org/10.11646/palaeoentomology.3.6.16>

Siphonaptera Literature (2021)

Alanazi, A. D., Alouffi, A. S., Alyousif, M. S., Alshahrani, M. Y., Abdullah, H. H. A. M., Abdel-Shafy, S., . . . Otranto, D. (2021). Molecular survey of vector-borne pathogens of dogs and cats in two regions of Saudi Arabia. Pathogens, 10(1), 1-11. doi:10.3390/pathogens10010025

Al-Doaissy, A. A., Al-Mekhlafi, F. A., Abutaha, N. M., Al-Keridis, L. A., Shati, A. A., Al-Kahtani, M. A., & Alfaihi, M. Y. (2021). Morphological, histological and ultrastructural characterisation of *Culex pipiens* (Diptera: Culicidae) larval midgut. African Entomology, 29(1), 274-288. doi:10.4001/003.029.0274

Al-Quraishy, S., Abdel-Ghaffar, F., Dkhil, M. A., & Abdel-Gaber, R. (2021). Haemogregarines and criteria for identification. Animals, 11(1), 1-25. doi:10.3390/ani11010170

Álvarez-Córdova, F., & Fernández, J. A. (2021). Medium-sized and large mammals of the pine-oak forests and grasslands of the Rancho Experimental Teseachi in Chihuahua, Mexico. Check List, 17(2), 669-681. doi:10.15560/17.2.669

Anderson, N. (2021). Safety concerns about the use of Seresto collars. Veterinary Record, 188(11), 437. doi:10.1002/vetr.620

Anstead, G. M. (2021). History, rats, fleas, and opossums. ii. The decline and resurgence of flea-borne typhus in the united states, 1945–2019. Tropical Medicine and Infectious Disease, 6(1) doi:10.3390/tropicalmed6010002

Antonovskaia, A. A., & Stekolnikov, A. A. (2021). Redescriptions of ten chigger mite species (Acariformes: Trombiculidae) from Vietnam. Zootaxa, 4969(1), 1-53. doi:10.11646/zootaxa.4969.1.1

Anyaele, O. O., & Enwemiwe, V. N. (2021). Prevalence of tungiasis in rural poor neighbourhood in Igbokoda, Ondo state, Nigeria. African Zoology, 56(2), 117-123. doi:10.1080/15627020.2021.1901604

Api, A. M., Belsito, D., Botelho, D., Bruze, M., Burton, G. A., Buschmann, J., . . . Tokura, Y. (2021). RIFM fragrance ingredient safety assessment, p-mentha-1,4-diene, CAS registry number 99-85-4. Food and Chemical Toxicology, 153 doi:10.1016/j.fct.2021.112359

Arias, A., Surugiu, V., Carballeira, R., Popa, O. P., Popa, L. O., & Utevsky, S. (2021). Unravelling the extent of diversity within the Iberian medicinal leeches (Hirudinea: Hirudo) using molecules and morphology. Biology, 10(4) doi:10.3390/biology10040315

- Azrizal-Wahid, N., Sofian-Azirun, M., & Low, V. L. (2021). Flea-borne pathogens in the cat flea *Ctenocephalides felis* and their association with mtDNA diversity of the flea host. *Comparative Immunology, Microbiology and Infectious Diseases*, 75 doi:10.1016/j.cimid.2021.101621
- Baardsen, L. F., De Bruyn, L., Adriaensen, F., Elst, J., Strubbe, D., Heylen, D., & Matthysen, E. (2021). No overall effect of urbanization on nest-dwelling arthropods of great tits (*Parus major*). *Urban Ecosystems*, doi:10.1007/s11252-020-01082-3
- Backus, L. H., López Pérez, A. M., & Foley, J. E. (2021). Effect of temperature on host preference in two lineages of the brown dog tick, *Rhipicephalus sanguineus*. *American Journal of Tropical Medicine and Hygiene*, 104(6), 2305-2311. doi:10.4269/ajtmh.20-1376
- Baláz, I., Ševčík, M., Tulis, F., Zigová, M., & Dudich, A. (2021). Diversity, distribution and changes in communities of fleas on small mammals along the elevational gradient from the Pannonian plain to the Carpathian mountains. *Parasitology*, 148(1), 63-73. doi:10.1017/S0031182020002024
- Baláz, I., Ševčík, M., Tulis, F., Zigová, M., & Dudich, A. (2021). Erratum: Diversity, distribution and changes in communities of fleas on small mammals along the elevational gradient from the Pannonian plain to the Carpathian mountains (*Parasitology* (2020) 148 (63-73) DOI: 10.1017/S0031182020002024). *Parasitology*, 148(5), 637. doi:10.1017/S0031182021000081
- Banda, A., Gandiwa, E., Muboko, N., & Muposhi, V. K. (2021). An assessment of rodent-flea diversity and association in a semi-arid tropical ecosystem of south-western Zimbabwe. *African Journal of Ecology*, doi:10.1111/aje.12867
- Barbieri, R., Signoli, M., Chevé, D., Costedoat, C., Tzortzis, S., Aboudharam, G., . . . Drancourt, M. (2021). *Yersinia pestis*: The natural history of plague. *Clinical Microbiology Reviews*, 34(1), 1-44. doi:10.1128/CMR.00044-19
- Barradas, P. F., Mesquita, J. R., Mateus, T. L., Ferreira, P., Amorim, I., Gärtner, F., & de Sousa, R. (2021). Molecular detection of *Rickettsia* spp. in ticks and fleas collected from rescued hedgehogs (*Erinaceus europaeus*) in Portugal. *Experimental and Applied Acarology*, 83(3), 449-460. doi:10.1007/s10493-021-00600-y
- Bartlett, J. C., Radcliffe, R. J., Convey, P., Hughes, K. A., & Hayward, S. A. L. (2021). The effectiveness of Virkon® S disinfectant against an invasive insect and implications for Antarctic biosecurity practices. *Antarctic Science*, 33(1), 1-9. doi:10.1017/S0954102020000413
- Bartlow, A. W., & Agosta, S. J. (2021). Phoresy in animals: Review and synthesis of a common but understudied mode of dispersal. *Biological Reviews*, 96(1), 223-246. doi:10.1111/brv.12654
- Bassini-Silva, R., Jacinavicius, F. D. C., Huang-Bastos, M., Dowling, A. P. G., & Barros-Battesti, D. M. (2021). A checklist of macronyssid species (Mesostigmata: Macronyssidae) from Brazil. *Journal of Medical Entomology*, 58(2), 625-633. doi:10.1093/jme/tjaa245
- Bäumer, W., & Baynes, R. (2021). Surface distribution of pyrethroids following topical application to veterinary species: Implications for lateral transport. *Journal of Veterinary Pharmacology and Therapeutics*, 44(1), 1-10. doi:10.1111/jvp.12907
- Beard, D., Stannard, H. J., & Old, J. M. (2021). Parasites of wombats (family Vombatidae), with a focus on ticks and tick-borne pathogens. *Parasitology Research*, 120(2), 395-409. doi:10.1007/s00436-020-07036-0

- Beliavskaia, A., Hönig, V., Erhart, J., Vyhldalova, T., Palus, M., Cerny, J., . . . Bell-Sakyi, L. (2021). *Spiroplasma* isolated from third-generation laboratory colony *Ixodes persulcatus* ticks. *Frontiers in Veterinary Science*, 8 doi:10.3389/fvets.2021.659786
- Berenbaum, M. R. (2021). Flea speech. *American Entomologist*, 67(2), 10-13. doi:10.1093/ae/tmab030
- Bernard, E. C., & Whittington, A. E. (2021). Papers and new species of minor insect orders published in *Zootaxa*, 2001–2020. *Zootaxa*, 4979(1), 232-235. doi:10.11646/zootaxa.4979.1.26
- Beutel, R. G., Friedrich, F., & Economo, E. P. (2021). Patterns of morphological simplification and innovation in the megadiverse Holometabola (Insecta). *Cladistics*, doi:10.1111/cla.12483
- Bezerra-Santos, M. A., Ramos, R. A. N., Campos, A. K., Dantas-Torres, F., & Otranto, D. (2021). *Didelphis* spp. opossums and their parasites in the Americas: A one health perspective. *Parasitology Research*, doi:10.1007/s00436-021-07072-4
- Bezerra-Santos, M. A., Sgroi, G., Mendoza-Roldan, J. A., Khedri, J., Camarda, A., Iatta, R., . . . Otranto, D. (2021). Ectoparasites of hedgehogs: From flea mite phoresy to their role as vectors of pathogens. *International Journal for Parasitology: Parasites and Wildlife*, 15, 95-104. doi:10.1016/j.ijppaw.2021.04.009
- Biggins, D. E., Eads, D. A., & Godbey, J. L. (2021). Plague transforms positive effects of precipitation on prairie dogs to negative effects. *International Journal for Parasitology: Parasites and Wildlife*, 14, 329-334. doi:10.1016/j.ijppaw.2021.02.006
- Biggins, D. E., Godbey, J. L., & Eads, D. A. (2021). Epizootic plague in prairie dogs: Correlates and control with Deltamethrin. *Vector-Borne and Zoonotic Diseases*, 21(3), 172-178. doi:10.1089/vbz.2020.2684
- Biggins, D. E., Ramakrishnan, S., Rocke, T. E., Williamson, J. L., & Wimsatt, J. (2021). Enzootic plague reduces survival of Mexican woodrats (*Neotoma mexicana*) in Colorado. *Ecosphere*, 12(2) doi:10.1002/ecs2.3371
- Bouvenot, T., Dewitte, A., Bennaceur, N., Pradel, E., Pierre, F., Bontemps-Gallo, S., & Sebbane, F. (2021). Interplay between *Yersinia pestis* and its flea vector in lipoate metabolism. *ISME Journal*, 15(4), 1136-1149. doi:10.1038/s41396-020-00839-0
- Brown, L. D., Maness, R., Hall, C., & Gibson, J. D. (2021). Reactive oxygen species-mediated immunity against bacterial infection in the gut of cat fleas (*Ctenocephalides felis*). *Insect Biochemistry and Molecular Biology*, 136 doi:10.1016/j.ibmb.2021.103620
- Buczek, A., & Buczek, W. (2021). Importation of ticks on companion animals and the risk of spread of tick-borne diseases to non-endemic regions in Europe. *Animals*, 11(1), 1-15. doi:10.3390/ani11010006
- Cermáková, P., Madarová, A., Baráth, P., Bellová, J., Yurchenko, V., & Horváth, A. (2021). Differences in mitochondrial NADH dehydrogenase activities in trypanosomatids. *Parasitology*, doi:10.1017/S0031182020002425
- Cevdanes, A., Ulloa-Contreras, C., Di Cataldo, S., Latrofa, M. S., Gonzalez-Acuña, D., Otranto, D., & Millán, J. (2021). Marked host association and molecular evidence of limited transmission of ticks and fleas between sympatric wild foxes and rural dogs. *Medical and Veterinary Entomology*, doi:10.1111/mve.12515

- Chaorattanakawee, S., Korkusol, A., Tippyachai, B., Promsathaporn, S., Poole-Smith, B. K., & Takhampunya, R. (2021). Amplicon-based next generation sequencing for rapid identification of *Rickettsia* and ectoparasite species from entomological surveillance in Thailand. *Pathogens*, 10(2), 1-11. doi:10.3390/pathogens10020215
- Colman, R. E., Brinkerhoff, R. J., Busch, J. D., Ray, C., Doyle, A., Sahl, J. W., . . . Wagner, D. M. (2021). No evidence for enzootic plague within black-tailed prairie dog (*Cynomys ludovicianus*) populations. *Integrative Zoology*, doi:10.1111/1749-4877.12546
- Cooke, B. D. (2021). Pasture plant biomass increase following introduction of European rabbit fleas, *Spilopsyllus cuniculi*, into Australia to facilitate myxomatosis transmission. *Biological Control*, 155 doi:10.1016/j.biocontrol.2021.104536
- Cosandey, V., Séchaud, R., Béziers, P., Chittaro, Y., Sanchez, A., & Roulin, A. (2021). Nidicolous beetle species richness is driven by barn owl's nests occupancy and landscape structure. *Journal of Ornithology*, 162(3), 857-864. doi:10.1007/s10336-021-01875-z
- Cruz-Bacab, L. E., Perez-De la Cruz, M. -, Zaragoza-Vera, C. V., Zaragoza-Vera, M., Arjona-Jimenez, G., Leshner-Gordillo, J. M., . . . Torres-Chable, O. M. (2021). Prevalence and factors associated with ectoparasite infestations in dogs from the state of Tabasco, Mexico. *The Journal of Parasitology*, 107(1), 29-38. doi:10.1645/20-71
- Dahm, J. R., Bailey, J. B., Kelly, R. F., Chikungwa, P., Chulu, J., Junior, L. C., . . . Sargison, N. D. (2021). Risk factors associated with *Ctenocephalides felis* flea infestation of peri-urban goats: A neglected parasite in an under-appreciated host. *Tropical Animal Health and Production*, 53(1) doi:10.1007/s11250-021-02620-7
- Dallas, T. A., & Becker, D. J. (2021). Taxonomic resolution affects host-parasite association model performance. *Parasitology*, 148(5), 584-590. doi:10.1017/S0031182020002371
- Danchenko, M., Laukaitis, H. J., & MacAluso, K. R. (2021). Dynamic gene expression in salivary glands of the cat flea during *Rickettsia felis* infection. *Pathogens and Disease*, 79(5) doi:10.1093/femspd/ftab020
- Dario, M. A., Lisboa, C. V., Silva, M. V., Herrera, H. M., Rocha, F. L., Furtado, M. C., . . . Jansen, A. M. (2021). *Crithidia mellificae* infection in different mammalian species in Brazil. *International Journal for Parasitology: Parasites and Wildlife*, 15, 58-69. doi:10.1016/j.ijppaw.2021.04.003
- Das, S. (2021). Taking a re-look at cap-binding signatures of the mRNA cap-binding protein eIF4E orthologues in trypanosomatids. *Molecular and Cellular Biochemistry*, 476(2), 1037-1049. doi:10.1007/s11010-020-03970-w
- de Araújo, F. S., Barcelos, R. M., Mendes, T. A. O., & Mafra, C. (2021). Molecular evidence of *Rickettsia felis* in *Phereoecca* sp. [Evidência molecular de rickettsia felis em phereoecca sp.] *Revista Brasileira De Parasitologia Veterinaria*, 30(1) doi:10.1590/S1984-29612021017
- Di Cataldo, S., Cevitanes, A., Ulloa-Contreras, C., Sacristán, I., Peñaloza-Madrid, D., Vianna, J., . . . Millán, J. (2021). Widespread infection with hemotropic mycoplasmas in free-ranging dogs and wild foxes across six bioclimatic regions of Chile. *Microorganisms*, 9(5) doi:10.3390/microorganisms9050919
- Douglas, G., Tsakris, A., Beleri, S., Patsoula, E., Linou, M., Billinis, C., & Papaparaskevas, J. (2021).

Molecular evidence of a broad range of pathogenic bacteria in *Ctenocephalides* spp.: Should we re-examine the role of fleas in the transmission of pathogens? *Tropical Medicine and Infectious Disease*, 6(1) doi:10.3390/tropicalmed6010037

Durães, L. S., Bitencourth, K., Ramalho, F. R., Nogueira, M. C., Nunes, E. D. C., & Gazêta, G. S. (2021). Biodiversity of potential vectors of rickettsiae and epidemiological mosaic of spotted fever in the state of Paraná, Brazil. *Frontiers in Public Health*, 9 doi:10.3389/fpubh.2021.577789

Durden, L. A., Bermúdez, S., Vargas, G. A., Sanjur, B. E., Gillen, L., Brown, L. D., . . . Eremeeva, M. E. (2021). Fleas (Siphonaptera) parasitizing peridomestic and indigenous mammals in Panamá and screening of selected fleas for vector-borne bacterial pathogens. *Journal of Medical Entomology*, 58(3), 1316-1321. doi:10.1093/jme/tjaa253

Dzul-Rosado, K. R., Reyes-Novelo, E., Lugo-Caballero, C., Cuxim-Koyoc, A. D., Collí-Padrón, F., Tello-Martín, R., . . . Ruiz-Piña, H. A. (2021). Urban ecology of hosts and vectors of *Rickettsia* in a rickettsiosis-endemic city of the Yucatan peninsula, Mexico. *Acta Tropica*, 216 doi:10.1016/j.actatropica.2021.105832

Eads, D. A., Livieri, T. M., Dobesh, P., Childers, E., Noble, L. E., Vasquez, M. C., & Biggins, D. E. (2021). Fipronil pellets reduce flea abundance on black-tailed prairie dogs: Potential tool for plague management and black-footed ferret conservation. *Journal of Wildlife Diseases*, 57(2), 434-438. doi:10.7589/JWD-D-20-00161

Eckerlin, R. P., & Gardner, A. L. (2021). Records of fleas (Siphonaptera) and other ectoparasites from Maine. *Northeastern Naturalist*, 28(1), 77-93. doi:10.1656/04028.0106

Eisen, R. J., Atiku, L. A., Ensore, R. E., Mpanga, J. T., Acayo, S., Mead, P. S., . . . Gage, K. L. (2021). Review article epidemiology, ecology and prevention of plague in the west Nile region of Uganda: The value of long-term field studies. *American Journal of Tropical Medicine and Hygiene*, 105(1), 18-23. doi:10.4269/ajtmh.20-1381

Enwemiwe, V. N., Ojianwuna, C. C., & Anyaele, O. O. (2021). Intensity and clinical morbidities of tungiasis in an impoverished south-west Nigerian community. *Parasite Epidemiology and Control*, 14 doi:10.1016/j.parepi.2021.e00215

Eren, G., Gürler, A. T., Gençay Topçu, E. B., Tuygun, T., & Açıcı, M. (2021). Türkiye’de bir kızıl tilkide (*vulpes vulpes*) felicola (*suricatoecus*) vulpis (phthiraptera: Trichodectidae) enfestasyonunun ilk bildirimi. [First Recorded *Felicola (Suricatoecus) vulpis* (Phthiraptera: Trichodectidae) Infestation in a Red Fox (*Vulpes vulpes*) in Turkey] *Türkiye Parazitolojii Dergisi*, 45(2), 157-159. doi:10.4274/tpd.galenos.2021.7047

Ezquiaga, M. C., Gallo, J. A., D'Agostino, R. L., Udrizar Sauthier, D. E., Abba, A. M., & Sanchez, J. (2021). Fleas and ticks in armadillos from Argentinean Patagonia: Diversity, abundance and distribution. *Acta Tropica*, 219 doi:10.1016/j.actatropica.2021.105911

Ezquiaga, M. C., & Soibelzon, E. (2021). New records of fleas and ticks in armadillos (Xenarthra: Chlamyphoridae) from the espinal and monte ecoregions of La Pampa province, Argentina. *Zootaxa*, 4950(3), 595-600. doi:10.11646/zootaxa.4950.3.12

Fagir, D. M., Bennett, N. C., Ueckermann, E. A., Howard, A., & Hart, D. W. (2021). Ectoparasitic community of the Mahali mole-rat, *Cryptomys hottentotus mahali*: Potential host for vectors of medical importance in south Africa. *Parasites and Vectors*, 14(1) doi:10.1186/s13071-020-04537-w

- Fang, K., Philpot, K., Chi, X., Ketzis, J., Du, A., & Yao, C. (2021). Small Indian mongooses (*Herpestes auropunctatus*) serve as reservoirs of *Bartonella henselae* and *Rickettsia felis* vectored by *Ctenocephalides felis*. *Vector-Borne and Zoonotic Diseases*, 21(6), 422-431. doi:10.1089/vbz.2020.2733
- Fenton, H., McBurney, S., Elsmo, E. J., Cleveland, C. A., & Yabsley, M. J. (2021). Lesions associated with *Bartonella taylorii*-like bacterium infection in a free-ranging, young-of-the-year raccoon from Prince Edward island, Canada. *Journal of Veterinary Diagnostic Investigation*, 33(2), 362-365. doi:10.1177/1040638720988515
- Fernandes, F. R., Da Silva Abreu, S., & Cruz, L. D. (2021). Transmission networks and ectoparasite mite burdens in *Oecomys paricola* (Rodentia: Cricetidae). *Parasitology*, 148(4), 443-450. doi:10.1017/S0031182020002231
- Flatau, R., Segoli, M., & Hawlena, H. (2021). *Wolbachia* endosymbionts of fleas occur in all females but rarely in males and do not show evidence of obligatory relationships, fitness effects, or sex-distorting manipulations. *Frontiers in Microbiology*, 12 doi:10.3389/fmicb.2021.649248
- Fried, G., Armengot, L., Storkey, J., Bourgeois, B., Gaba, S., Violle, C., & Munoz, F. (2021). Do ecological specialization and functional traits explain the abundance–frequency relationship? arable weeds as a case study. *Journal of Biogeography*, 48(1), 37-50. doi:10.1111/jbi.13980
- Frolov, A. O., Kostygov, A. Y., & Yurchenko, V. (2021). Development of monoxenous trypanosomatids and phytomonads in insects. *Trends in Parasitology*, 37(6), 538-551. doi:10.1016/j.pt.2021.02.004
- Fuehrer, H. -, Soukup, I., Shahi-Barogh, B., & Glawischnig, W. (2021). Absence of blood parasites and other vector-borne pathogens in alpine marmots (*Marmota marmota*) in western Austria. *Parasitology Research*, 120(3), 1125-1129. doi:10.1007/s00436-020-07006-6
- Furquim, M. E. C., do Amaral, R., Dias, C. M., Gonçalves, L. R., Perles, L., Lima, C. A. D. P., . . . André, M. R. (2021). Genetic diversity and multilocus sequence typing analysis of *Bartonella henselae* in domestic cats from southeastern Brazil. *Acta Tropica*, 222 doi:10.1016/j.actatropica.2021.106037
- Galán-Puchades, M. T., Mas-Coma, S., Valero, M. A., & Fuentes, M. V. (2021). First data on the helminth community of the smallest living mammal on earth, the Etruscan pygmy shrew, *Suncus etruscus* (Savi, 1822) (Eulipotyphla: Soricidae). *Animals*, 11(7) doi:10.3390/ani11072074
- Galloway, T. D., & Lamb, R. J. (2021). Population dynamics of chewing lice (Phthiraptera) infesting birds (Aves) doi:10.1146/annurev-ento-041420-075608
- García-Bocanegra, I., Camacho-Sillero, L., Caballero-Gómez, J., Agüero, M., Gómez-Guillamón, F., Manuel Ruiz-Casas, J., . . . de la Haza, R. (2021). Monitoring of emerging *Myxoma* virus epidemics in Iberian hares (*Lepus granatensis*) in Spain, 2018–2020. *Transboundary and Emerging Diseases*, 68(3), 1275-1282. doi:10.1111/tbed.13781
- Geary, M. J., Russell, R. C., Moerkerken, L., Hassan, A., & Doggett, S. L. (2021). 30 years of samples submitted to an Australian medical entomology department. *Austral Entomology*, 60(1), 172-197. doi:10.1111/aen.12480
- Goldberg, A. R., Conway, C. J., & Biggins, D. E. (2021). Effects of experimental flea removal and plague vaccine treatments on survival of northern idaho ground squirrels and two coexisting sciurids. *Global Ecology and Conservation*, 26 doi:10.1016/j.gecco.2021.e01489

- Gómez-Undiano, I., & París, M. (2021). Catalogue of mecopterans (Insecta, Mecoptera) in the museo nacional de ciencias naturales (csic). [Catálogo de los mecópteros (Insecta, mecoptera) del museo nacional de ciencias naturales (csic)] Graellsia, 77(1) doi:10.3989/GRAELLSIA.2021.V77.269
- Gong, C., Wang, X., Huang, Q., Zhang, J., Zhang, Y., Zhan, X., . . . Shen, L. (2021). The fitness advantages of Bistrifluron resistance related to chitin synthase A in *Spodoptera litura* (Fab.) (Noctuidae: Lepidoptera). Pest Management Science, 77(7), 3458-3468. doi:10.1002/ps.6399
- Graciolli, G., Ávila-Palma, H. D., Ordoñez-Trejo, E. J., Soler-Orellana, J. A., Ordoñez-Mazier, D. I., Martínez, M., . . . Turcios-Casco, M. A. (2021). Additions of host associations and new records of bat ectoparasites of the families Spinturnicidae, Nycteribiidae and Streblidae from honduras. Check List, 17(2), 459-469. doi:10.15560/17.2.459
- Hamidi, K., & Bueno-Marí, R. (2021). Host-ectoparasite associations; the role of host traits, season and habitat on parasitism interactions of the rodents of northeastern Iran. Journal of Asia-Pacific Entomology, 24(1), 308-319. doi:10.1016/j.aspen.2020.12.009
- Hamidi, K., Hamidi, A., & Bueno-Marí, R. (2021). Potential infections and zoonotic diseases transmitted by small laboratory mammals. Advances in health and disease (pp. 111-142) Retrieved from www.scopus.com
- Harvey, T. V., dos Santos Freire, Z., dos Santos, K. C., de Jesus, A. V., Guedes, P. E. B., da Paixão Sevá, A., . . . Carlos, R. S. A. (2021). Clinical and macroscopic morphological features of canine tungiasis. Parasitology Research, 120(3), 807-818. doi:10.1007/s00436-020-07013-7
- Hastriter, M. W. (2021). Fleas (Siphonaptera: Pygiopsyllomorpha) of Papua New Guinea and Papua province (Indonesia). Part VI. *Bibikovana*, *Geohollandia*, and *Hoogstraalia* (Pygiopsyllidae: Pygiopsyllinae), with descriptions of four new species. Annals of Carnegie Museum, 87(1), 37-77. doi:10.2992/007.087.0102
- Heath, A. C. G. (2021). Climate change and its potential for altering the phenology and ecology of some common and widespread arthropod parasites in New Zealand. New Zealand Veterinary Journal, 69(1), 5-19. doi:10.1080/00480169.2020.1787276
- Henriquez, B. E., Featherston, K., & Noden, B. H. (2021). Detection of *Bartonella henselae* and *Rickettsia felis* in fleas from client-owned pets in Oklahoma. Southwestern Entomologist, 46(1), 253-255. doi:10.3958/059.046.0124
- Herrero-Cófreces, S., Flechoso, M. F., Rodríguez-Pastor, R., Luque-Larena, J. J., & Mougeot, F. (2021). Patterns of flea infestation in rodents and insectivores from intensified agro-ecosystems, northwest Spain. Parasites and Vectors, 14(1) doi:10.1186/s13071-020-04492-6
- Himsworth, C. G., Byers, K. A., Whelan, T., Bai, Y., & Kosoy, M. Y. (2021). Flea presence and abundance are not predictors of *Bartonella tribocorum* carriage in Norway rats *Rattus norvegicus* from an underserved neighborhood of Vancouver, Canada. Vector-Borne and Zoonotic Diseases, 21(2), 121-124. doi:10.1089/vbz.2020.2665
- Hinds, L. A., Grice, D., Watson, D. M., & Jacob, J. (2021). Efficacy of a combined insecticide-rodenticide product on ectoparasite and commensal rodent mortality. Pest Management Science, 77(3), 1160-1168. doi:10.1002/ps.6179
- Hinnebusch, B. J., Jarrett, C. O., & Bland, D. M. (2021). Molecular and genetic mechanisms that

- mediate transmission of *Yersinia pestis* by fleas. *Biomolecules*, 11(2), 1-13. doi:10.3390/biom11020210
- Ho, J., Changbunjong, T., Weluwanarak, T., Hussain, S., & Sparagano, O. (2021). The pests of a pest: A systematic review of ectoparasitic fauna among synanthropic rodents in the 21st century with meta-analysis. *Acta Tropica*, 215 doi:10.1016/j.actatropica.2020.105802
- Hu, L., Zhao, Y., Yang, Y., Zhang, W., Guo, H., & Niu, D. (2021). Molecular identification, transcriptome sequencing and functional annotation of *Pulex irritans*. *Acta Parasitologica*, 66(2), 605-614. doi:10.1007/s11686-020-00296-x
- Huggins, L. G., Colella, V., Koehler, A. V., Schunack, B., & Traub, R. J. (2021). A multipronged next-generation sequencing metabarcoding approach unearths hyperdiverse and abundant dog pathogen communities in Cambodia. *Transboundary and Emerging Diseases*, doi:10.1111/tbed.14180
- Islam, M. M., Farag, E., Eltom, K., Hassan, M. M., Bansal, D., Schaffner, F., . . . Mkhize-Kwitshana, Z. (2021). Rodent ectoparasites in the middle East: A systematic review and meta-analysis. *Pathogens*, 10(2), 1-22. doi:10.3390/pathogens10020139
- Jacob, J., Aplin, K., Watson, D. M., & Hinds, L. A. (2021). Assessing the efficacy of oral intake of insecticides on mortality of fleas and ticks on commensal *Rattus* species. *Journal of Pest Science*, doi:10.1007/s10340-020-01316-5
- Jaszayova, A., & Jaszay, T. (2021). Pseudoscorpions (Arachnida: Pseudoscorpiones) from leaf litter of the Slovak karst national park. *Arachnologische Mitteilungen*, 61(1), 77-83. doi:10.30963/aramit6113
- Jiang, J., Farris, C. M., Yeh, K. B., & Richards, A. L. (2021). International *Rickettsia* disease surveillance: An example of cooperative research to increase laboratory capability and capacity for risk assessment of rickettsial outbreaks worldwide. *Frontiers in Medicine*, 8 doi:10.3389/fmed.2021.622015
- Kasozzi, K. I., Zirintunda, G., Ssempijja, F., Buyinza, B., Alzahrani, K. J., Matama, K., . . . Welburn, S. C. (2021). Epidemiology of trypanosomiasis in Wildlife—Implications for humans at the wildlife interface in Africa. *Frontiers in Veterinary Science*, 8 doi:10.3389/fvets.2021.621699
- Kerr, P. J., Hall, R. N., & Strive, T. (2021). Viruses for landscape-scale therapy: Biological control of rabbits in Australia doi:10.1007/978-1-0716-1012-1_1
- Keskin, A. (2021). Description of a new flea species of the genus *Ctenophthalmus* kolenati (siphonaptera: Ctenophthalmidae) from Turkey. *Parasitology Research*, 120(4), 1203-1207. doi:10.1007/s00436-021-07076-0
- Khokhlova, I. S., van der Mescht, L., Warburton, E. M., Stavtseva, N. A., & Krasnov, B. R. (2021). Adaptation to a novel host and performance trade-off in host-generalist and host-specific insect ectoparasites. *Insect Science*, doi:10.1111/1744-7917.12929
- Kiene, F., Andriatsitohaina, B., Ramsay, M. S., Rakotondravony, R., Strube, C., & Radespiel, U. (2021). Habitat fragmentation and vegetation structure impact gastrointestinal parasites of small mammalian hosts in Madagascar. *Ecology and Evolution*, 11(11), 6766-6788. doi:10.1002/ece3.7526
- Kinderlehrer, D. A. (2021). Anorexia nervosa caused by polymicrobial tick-borne infections: A case study. *International Medical Case Reports Journal*, 14, 279-287. doi:10.2147/IMCRJ.S311516
- Kirillova, N. Y., Kirillov, A. A., & Ruchin, A. B. (2021). First record of helminths of the european pine vole, *Microtus subterraneus* (Rodentia, Cricetidae) in Russia with overview on the rodent's

range. Russian Journal of Theriology, 20(1), 19-24. doi:10.15298/RUSJTHERIOL.20.1.03

Kissayi, K., Mitroiu, M. -, & Rohi, L. (2021). Annotated check-list of Pteromalidae (Hymenoptera: Chalcidoidea) of Morocco. Part ii. [Lista comentada de pteromalidae (Hymenoptera: chalcidoidea) de marruecos. parte ii] Graellsia, 77(1) doi:10.3989/GRAELLSIA.2021.V77.301

Kłys, G., & Lis, B. (2021). Living on the edge: *Meoneura obscurella* in the ‘Wieliczka’ salt mine (southern Poland) exhibits the first case of lecithotrophic ovoviviparity in the family Carnidae (Diptera). European Zoological Journal, 88(1), 466-471. doi:10.1080/24750263.2021.1899315

Kostygov, A. Y., Karnkowska, A., Votýpka, J., Tashyreva, D., Maciszewski, K., Yurchenko, V., & Lukeš, J. (2021). Euglenozoa: Taxonomy, diversity and ecology, symbioses and viruses. Open Biology, 11(3) doi:10.1098/rsob.200407

Krasnov, B. R., Spickett, A., Junker, K., Bugmyrin, S. V., Ieshko, E. P., Bespyatova, L. A., . . . Matthee, S. (2021). Parasite counts or parasite incidences? Testing differences with four analyses of infracommunity modelling for seven parasite–host associations. Parasitology Research, 120(7), 2569-2584. doi:10.1007/s00436-021-07217-5

Krasnov, B. R., Spickett, A., Junker, K., van der Mescht, L., & Matthee, S. (2021). Functional and phylogenetic uniqueness of helminth and flea assemblages of two South African rodents. International Journal for Parasitology, doi:10.1016/j.ijpara.2021.02.003

Krasnov, B. R., Vinarski, M. V., Korralo-Vinarskaya, N. P., Shenbrot, G. I., & Khokhlova, I. S. (2021). Spatial and temporal variation of compositional, functional, and phylogenetic diversity in ectoparasite infracommunities harboured by small mammals. Parasitology, 148(6), 685-695. doi:10.1017/S0031182021000299

Landaeta-Aqueveque, C., Moreno Salas, L., Henríquez, A., Silva-de la Fuente, M. C., & González-Acuña, D. (2021). Parasites of native and invasive rodents in Chile: Ecological and human health needs. Frontiers in Veterinary Science, 8 doi:10.3389/fvets.2021.643742

Laukaitis, H. J., & Macaluso, K. R. (2021). Unpacking the intricacies of *Rickettsia*–vector interactions. Trends in Parasitology, 37(8), 734-746. doi:10.1016/j.pt.2021.05.008

Lehikoinen, A., Pohjola, P., Valkama, J., Mutanen, M., & Pohjoismäki, J. L. O. (2021). Promiscuous specialists: Host specificity patterns among generalist louse flies. PLoS ONE, 16(5 May) doi:10.1371/journal.pone.0247698

Lian, X., Cai, C., & Huang, D. (2021). Revision of the long-proboscid scorpionflies, *Lichnomesopsyche* ren, labandeira, and shih, 2010. Journal of Paleontology, 95(4), 793-799. doi:10.1017/jpa.2021.21

Lima-Duarte, L., Camargo, J. V., Castro-Santiago, A. C., Machado, R. Z., André, M. R., Cabral-de-Mello, D. C., . . . Barros-Battesti, D. M. (2021). Establishment and characterization of a cell line (RBME-6) of *Rhipicephalus (Boophilus) microplus* from brazil. Ticks and Tick-Borne Diseases, 12(5) doi:10.1016/j.ttbdis.2021.101770

López-Pérez, A. M., Chaves, A., Sánchez-Montes, S., Foley, P., Uhart, M., Barrón-Rodríguez, J., . . . Foley, J. (2021). Diversity of rickettsiae in domestic, synanthropic, and sylvatic mammals and their ectoparasites in a spotted fever-epidemic region at the western US-Mexico border. Transboundary and Emerging Diseases, doi:10.1111/tbed.14027

- Lukeš, J., Tesařová, M., Yurchenko, V., & Votýpka, J. (2021). Characterization of a new cosmopolitan genus of trypanosomatid parasites, *Obscuromonas* gen. nov. (bBastocrithidiinae subfam. nov.). *European Journal of Protistology*, 79 doi:10.1016/j.ejop.2021.125778
- Ma, N., Zhang, Y. -, & Yue, C. (2021). The histology and ultrastructure of the salivary glands of *Neopanorpa longiprocessa* (Mecoptera: Panorpidae). *Protoplasma*, 258(1), 59-69. doi:10.1007/s00709-020-01549-2
- Machač, O., & Tuf, I. H. (2021). Ornithologists' help to spiders: Factors influencing spiders overwintering in bird nesting boxes. *Insects*, 12(5) doi:10.3390/insects12050465
- Magri, A., Galuppi, R., & Fioravanti, M. (2021). Autochthonous *Trypanosoma* spp. in European mammals: A brief journey amongst the neglected trypanosomes. *Pathogens*, 10(3) doi:10.3390/pathogens10030334
- Malloy, G. S. P., Brandeau, M. L., & Goldhaber-Fiebert, J. D. (2021). Modeling the cost-effectiveness of interventions to prevent plague in Madagascar. *Tropical Medicine and Infectious Disease*, 6(2) doi:10.3390/tropicalmed6020101
- Manoj, R. R. S., Latrofa, M. S., Epis, S., & Otranto, D. (2021). *Wolbachia*: Endosymbiont of Onchocercid nematodes and their vectors. *Parasites and Vectors*, 14(1) doi:10.1186/s13071-021-04742-1
- Marchenko, I. I. (2021). Four new species of *Halozercon* (Acari: Mesostigmata: Zerconidae) from south Siberia mountains (Russia) with a key to all known species. *Zootaxa*, 4941(2), 151-185. doi:10.11646/zootaxa.4941.2.1
- Martínez-Díaz, H. -, Forero-Becerra, E., Hidalgo, M., & Labruna, M. B. (2021). Experimental infection and vector competence of *Amblyomma patinoi*, a member of the *Amblyomma cajennense* species complex, for the human pathogen *Rickettsia rickettsii*. *Ticks and Tick-Borne Diseases*, 12(5) doi:10.1016/j.ttbdis.2021.101751
- Maruyama, S. R., Rogerio, L. A., Freitas, P. D., Teixeira, M. M. G., & Ribeiro, J. M. C. (2021). Total ortholog median matrix as an alternative unsupervised approach for phylogenomics based on evolutionary distance between protein coding genes. *Scientific Reports*, 11(1) doi:10.1038/s41598-021-81926-w
- Mau, A., Calchi, A. C., Bittencourt, P., Navarrete-Talloni, M. J., Sauvé, C., Conan, A., . . . Müller, A. (2021). Molecular survey and genetic diversity of *Bartonella* spp. in small Indian mongooses (*Urva auropunctata*) and their fleas on Saint Kitts, West Indies. *Microorganisms*, 9(7) doi:10.3390/microorganisms9071350
- Mayorga, C. G. A., Tello, G. B., González, I. A., Vallejo, D. A. Z., & Novoa, M. S. (2021). *Cavia porcellus*, reservoir of *Yersinia pestis* in the border areas Peru - Ecuador: Education is the key to control. [*Cavia porcellus*, reservorio del *Yersinia pestis* en los focos fronterizos Perú - Ecuador: Educación la clave en el control] *Boletín De Malariología y Salud Ambiental*, 61, 14-21. doi:10.52808/bmsa.7e5.61e.002
- McKee, C. D., Bai, Y., Webb, C. T., & Kosoy, M. Y. (2021). Bats are key hosts in the radiation of mammal-associated *Bartonella* bacteria. *Infection, Genetics and Evolution*, 89 doi:10.1016/j.meegid.2021.104719

- McLaughlin, L. G., & Wasserberg, G. (2021). Spatial bet hedging in sand fly oviposition: Factors affecting skip oviposition in *Phlebotomus papatasi* sand flies. *Vector-Borne and Zoonotic Diseases*, 21(4), 280-288. doi:10.1089/vbz.2020.2737
- Miarinjara, A., Bland, D. M., Belthoff, J. R., & Hinnebusch, B. J. (2021). Poor vector competence of the human flea, *Pulex irritans*, to transmit *Yersinia pestis*. *Parasites and Vectors*, 14(1) doi:10.1186/s13071-021-04805-3
- Michalko, R., & Birkhofer, K. (2021). Habitat niches suggest that non-crop habitat types differ in quality as source habitats for central European agrobiont spiders. *Agriculture, Ecosystems and Environment*, 308 doi:10.1016/j.agee.2020.107248
- Milholland, M. T., Eisen, L., Nadolny, R. M., Hojgaard, A., Machtinger, E. T., Mullinax, J. M., & Li, A. Y. (2021). Surveillance of ticks and tick-borne pathogens in suburban natural habitats of central Maryland. *Journal of Medical Entomology*, 58(3), 1352-1362. doi:10.1093/jme/tjaa291
- Minaya, D., Mendoza, J., & Iannaccone, J. (2021). Ectoparasitic fauna on the common vampire bat *Desmodus rotundus* (Geoffroy, 1810) (Chiroptera: Phyllostomidae) from Huarochiri, Lima, and a checklist of ectoparasites in bats of Peru. [Fauna de ectoparásitos en el vampiro común *Desmodus rotundus* (Geoffroy, 1810) (Chiroptera: Phyllostomidae) de Huarochiri, Lima, y una lista de los ectoparásitos en murciélagos del Perú] *Graellsia*, 77(1) doi:10.3989/graeellsia.2021.v77.293
- Mohammadi, S., Lutermann, H., Hoffmann, S., Emami-Khoyi, A., Webster, H. J., . . . van Vuuren, B. J. 2021. MORPHOLOGICAL AND MOLECULAR CHARACTERIZATION OF THE PLAGUE VECTOR *XENOPSYLLA BRASILIENSIS*. *J. Parasitol.* 107(2):289-294. doi:10.1645/20-44
- Moudgil, A. D., & Das Singla, L. (2021). Scanning electron microscopy based identification of ectoparasitic insects infesting zoo-housed jungle cats and pigeons of north India. *International Journal of Tropical Insect Science*, doi:10.1007/s42690-021-00458-y
- Moura, J. F., Nascimento, C. S. I., Peixoto, B. D. C. P. E. M., de Barros, G. E. B., Robbi, B., & Fernandes, M. A. (2021). Damaged armour: Ichnotaxonomy and paleoparasitology of bioerosion lesions in osteoderms of Quaternary extinct armadillos. *Journal of South American Earth Sciences*, 109 doi:10.1016/j.jsames.2021.103255
- Mutebi, F., Krücken, J., Feldmeier, H., & von Samsom-Himmelstjerna, G. 2021. Clinical implications and treatment options of tungiasis in domestic animals. *Parasitol. Res.*doi:10.1007/s00436-021-07121-y
- Myczko, Ł., Kurek, P., Tryjanowski, P., Wiatrowska, B., Jankowiak, Ł., Mielczarek, Ł., . . . Ondrejková, A. (2021). Where to overwinter: Burrows of medium-sized carnivores as winter places for invertebrates in temperate environment. *Ecological Entomology*, doi:10.1111/een.13062
- Namaki-Khameneh, R., Khaghaninia, S., Disney, R. H. L., & Maleki-Ravasan, N. (2021). Nine new species of scuttle flies, including one new genus (Diptera: Phoridae) from Iran. *Biologia*, doi:10.1007/s11756-021-00762-5
- Nangoy, M., Ransaleleh, T., Lengkong, H., Koneri, R., Latinne, A., & Kyes, R. C. (2021). Diversity of fruit bats (Pteropodidae) and their ectoparasites in Batuputih nature tourism park, Sulawesi, Indonesia. *Biodiversitas*, 22(6), 3075-3082. doi:10.13057/biodiv/d220609
- Napierała, A., Maziarz, M., Hebda, G., Broughton, R. K., Rutkowski, T., Zacharyasiewicz, M., & Błoszyk, J. (2021). Lack of specialist nidicoles as a characteristic of mite assemblages inhabiting nests

- of the ground-nesting wood warbler, *Phylloscopus sibilatrix* (Aves: Passeriformes). *Experimental and Applied Acarology*, 84(1), 149-170. doi:10.1007/s10493-021-00620-8
- Nuhličková, S., Svetlík, J., Eckenfellner, M., Knauer, F., & Hoi, H. (2021). Interaction between nestling behaviour and nest-space use. *Ethology Ecology and Evolution*, doi:10.1080/03949370.2020.1858173
- Obdianela, M. C. N., Guanlao, M. P. V. V., Samaniego, E. V. E., & Pornobi, K. O. (2021). Prevalence and host specificity of bat flies (Streblidae) from selected caves in Unisan, Quezon, Philippines. *Acta Parasitologica*, doi:10.1007/s11686-021-00373-9
- Obiegala, A., Arnold, L., Pfeffer, M., Kiefer, M., Kiefer, D., Sauter-Louis, C., & Silaghi, C. (2021). Host–parasite interactions of rodent hosts and ectoparasite communities from different habitats in Germany. *Parasites and Vectors*, 14(1) doi:10.1186/s13071-021-04615-7
- Obiegala, A., Pfeffer, M., Kiefer, D., Kiefer, M., Król, N., & Silaghi, C. (2021). *Bartonella* spp. in small mammals and their fleas in differently structured habitats from Germany. *Frontiers in Veterinary Science*, 7 doi:10.3389/fvets.2020.625641
- Occi, J. L., Hall, M., Egizi, A. M., Robbins, R. G., & Fonseca, D. M. (2021). First record of *Carios kelleyi* (Acari: Ixodida: Argasidae) in New Jersey, United States and implications for public health. *Journal of Medical Entomology*, 58(2), 939-942. doi:10.1093/jme/tjaa189
- Ogata, S., Mohamed, W. M. A., Kusakisako, K., Thu, M. J., Qiu, Y., Moustafa, M. A. M., . . . Nakao, R. (2021). Article *Spiroplasma* infection among ixodid ticks exhibits species dependence and suggests a vertical pattern of transmission. *Microorganisms*, 9(2), 1-17. doi:10.3390/microorganisms9020333
- Oglezneva, A. A., Durnova, N. A., & Voronin, M. Y. (2021). Populational and karyological analysis of the phytophilous chironomid *Endochironomus tendens* F. (Diptera, Chironomidae). 2. Chromosomal polymorphism and cytogenetic differentiation in populations. *Entomological Review*, 101(1), 59-70. doi:10.1134/S001387382101005X
- Ohnishi, O., Takenaka, M., Okano, R., Yoshitomi, H., & Tojo, K. (2021). Wide-scale gene flow, even in insects that have lost their flight ability: Presence of dispersion due to a unique parasitic ecological strategy of piggybacking hosts. *Zoological Science*, 38(2), 122-139. doi:10.2108/zs200088
- Okaro, U., George, S., & Anderson, B. (2021). What is in a cat scratch? Growth of *Bartonella henselae* in a biofilm. *Microorganisms*, 9(4) doi:10.3390/microorganisms9040835
- Oliveira, J. C. P., Oliveira, W. S. M., Brito, R. S., Lima, T. A. R. F., Giannelli, A., Carvalho, G. A., & Ramos, R. A. N. (2021). Ectoparasites infesting animals living in close contact with human beings: A real trouble for one health perspective? [Ectoparasitos infestando animais que vivem em contato próximo com seres humanos: um problema real para a perspectiva Saúde Única?] *Arquivo Brasileiro De Medicina Veterinaria e Zootecnia*, 73(1), 55-61. doi:10.1590/1678-4162-12084
- Orlova, M. V., Klimov, P. B., Moskvitina, N. S., Orlov, O. L., Zhigalin, A. V., Smirnov, D. G., . . . Ekaterina Khristenko. (2021). New records of bat flies (Diptera: Nycteribiidae), with an updated checklist of the nycteribiids of Russia. *Zootaxa*, 4927(3), 410-430. doi:10.11646/zootaxa.4927.3.5
- Orlova, M. V., Klimov, P. B., Orlov, O. L., Krusko, S. V., & Lebedev, V. S. (2021). New geographic and host records of spinturnicid mites (Mesostigmata: Spinturnicidae) in Asia, with description of the protonymph of *Spinturnix tylonycterisi*. *International Journal of Acarology*, 47(4), 361-365. doi:10.1080/01647954.2021.1912177

- Orlova, M. V., Klimov, P. B., Orlov, O. L., Smirnov, D. G., Zhigalin, A. V., Budaeva, I. V., . . . Anisimov, N. V. (2021). A checklist of bat-associated macronyssid mites (Acari: Gamasina: Macronyssidae) of Russia, with new host and geographical records. *Zootaxa*, 4974(3), 537-564. doi:10.11646/zootaxa.4974.3.4
- Osikowicz, L. M., Horiuchi, K., Goodrich, I., Breitschwerdt, E. B., Chomel, B., Biggerstaff, B. J., & Kosoy, M. (2021). Exposure of domestic cats to three zoonotic *Bartonella* species in the united states. *Pathogens*, 10(3) doi:10.3390/pathogens10030354
- Osorio-Pinzon, J., Palencia, A., Cruz-Calderon, S., & Rodriguez-Morales, A. J. (2021). Myiasis and tungiasis. *Current Tropical Medicine Reports*, 8(2), 112-120. doi:10.1007/s40475-021-00233-6
- Otiang, E., Chen, D., Jiang, J., Maina, A. N., Farris, C. M., Luce-Fedrow, A., & Richards, A. L. (2021). Pathogen carriage by peri-domestic fleas in western Kenya. *Vector-Borne and Zoonotic Diseases*, 21(4), 256-263. doi:10.1089/vbz.2020.2709
- Otranto, D., Dantas-Torres, F., Fourie, J. J., Lorusso, V., Varlout, M., Gradoni, L., . . . Holdsworth, P. (2021). World association for the advancement of veterinary parasitology (W.A.A.V.P.) guidelines for studies evaluating the efficacy of parasiticides in reducing the risk of vector-borne pathogen transmission in dogs and cats. *Veterinary Parasitology*, 290 doi:10.1016/j.vetpar.2021.109369
- Ouarti, B., Hamzaoui, B. E., Stanko, M., Laroche, M., Mediannikov, O., Parola, P., & Sekeyová, Z. (2021). Detection of *Rickettsia raoultii* in *Dermacentor reticulatus* and *Haemaphysalis inermis* ticks in Slovakia. *Biologia*, doi:10.1007/s11756-021-00789-8
- Pacifico, L., Ferrari, N., Romeo, C., Buono, F., Varuzza, P., Sgroi, G., . . . Piantedosi, D. (2021). Haematological and biochemical abnormalities in hunting dogs infected with *Acanthocheilonema reconditum*, associated risk factors, and a European overview. *Parasitology Research*, 120(6), 2109-2124. doi:10.1007/s00436-021-07179-8
- Panthawong, A., Doggett, S. L., & Chareonviriyaphap, T. (2021). The efficacy of ultrasonic pest repellent devices against the Australian paralysis tick, *Ixodes holocyclus* (Acari: Ixodidae). *Insects*, 12(5) doi:10.3390/insects12050400
- Perkins, R., Whitehead, M., Civil, W., & Goulson, D. (2021). Potential role of veterinary flea products in widespread pesticide contamination of English rivers. *Science of the Total Environment*, 755 doi:10.1016/j.scitotenv.2020.143560
- Péter, Á., Mihalca, A. D., & Sándor, A. D. (2021). First report of the bat fly species *Basilia italica* in Romania. *Biodiversity Data Journal*, 9(2), 1-7. doi:10.3897/BDJ.9.E57680
- Philip Samuel, P., Govindarajan, R., Krishnamoorthi, R., & Nagaraj, J. (2021). Ectoparasites of some wild rodents /shrews captured from scrub typhus reported areas in Tamil Nadu, India. *International Journal of Acarology*, 47(3), 218-221. doi:10.1080/01647954.2021.1887932
- Porter, M. L., Cronin, T. W., Dick, C. W., Simon, N., & Dittmar, K. (2021). Visual system characterization of the obligate bat ectoparasite *Trichobius frequens* (Diptera: Streblidae). *Arthropod Structure and Development*, 60 doi:10.1016/j.asd.2020.101007
- Pyszek, P., Plášek, V., & Drozd, P. (2021). Don't eat where you sleep: Unexpected diversity of food web for beetles feeding on mosses. *Insect Conservation and Diversity*, 14(3), 325-334. doi:10.1111/icad.12453

- Rahelinirina, S., Scobie, K., Ramasindrazana, B., Andrianaiivoarimanana, V., Rasoamalala, F., Randrianitseho, L. N., . . . Rajerison, M. (2021). Rodent control to fight plague: Field assessment of methods based on rat density reduction. *Integrative Zoology*, doi:10.1111/1749-4877.12529
- Rolbiecki, L., Izdebska, J. N., Franke, M., Iliszko, L., & Fryderyk, S. (2021). The vector-borne zoonotic nematode *Thelazia callipaeda* in the eastern part of Europe, with a clinical case report in a dog in Poland. *Pathogens*, 10(1), 1-7. doi:10.3390/pathogens10010055
- Rosche, K. L., Sidak-Loftis, L. C., Hurtado, J., Fisk, E. A., & Shaw, D. K. (2021). Arthropods under pressure: Stress responses and immunity at the pathogen-vector interface. *Frontiers in Immunology*, 11 doi:10.3389/fimmu.2020.629777
- Rosenzweig, J. A., Hendrix, E. K., & Chopra, A. K. (2021). Plague vaccines: New developments in an ongoing search. *Appl. Microbiol. Biotech.*, 105(12), 4931-4941. doi:10.1007/s00253-021-11389-6
- Rukke, B. A., Salma, U., Birkemoe, T., & Aak, A. (2021). Blood deprivation and heat stress increase mortality in bed bugs (*Cimex lectularius*) exposed to insect pathogenic fungi or desiccant dust. *Med. Vet. Entomol.*, 35(1), 121-128. doi:10.1111/mve.12477
- Russell, R. E., Walsh, D. P., Samuel, M. D., Grunnill, M. D., & Rocke, T. E. (2021). Space matters: Host spatial structure and the dynamics of plague transmission. *Ecological Modelling*, 443 doi:10.1016/j.ecolmodel.2021.109450
- Saengsawang, P., Kaewmongkol, G., & Inpankaew, T. (2021). Molecular detection of *Bartonella* spp. and hematological evaluation in domestic cats and dogs from Bangkok, Thailand. *Pathogens*, 10(5) doi:10.3390/pathogens10050503
- Saichenko, I. V., Antipov, A. A., Bakhur, T. I., Bezditzko, L. V., & Shmayun, S. S. (2021). Co-infection of *Trichuris vulpis* and *Toxocara canis* in different aged dogs: Influence on the haematological indices. *Biosystems Diversity*, 29(2), 129-134. doi:10.15421/012117
- Sánchez-Montes, S., Colunga-Salas, P., Lozano-Sardaneta, Y. N., Zazueta-Islas, H. M., Ballados-González, G. G., Salceda-Sánchez, B., . . . Becker, I. (2021). The genus *Rickettsia* in Mexico: Current knowledge and perspectives. *Ticks and Tick-Borne Diseases*, 12(2) doi:10.1016/j.ttbdis.2020.101633
- Santoyo-Colín, V., Sánchez-Montes, S., Salceda-Sánchez, B., Huerta-Jiménez, H., Alcántara-Rodríguez, V., Becker, I., . . . López-Pérez, A. M. (2021). Urban foci of murine typhus involving cat fleas (*Ctenocephalides felis felis*) collected from opossums in Mexico city. *Zoonoses and Public Health*, 68(1), 1-7. doi:10.1111/zph.12770
- Savassi, B. A. E. S., Dobigny, G., Etougbétché, J. R., Avocegan, T. T., Quinsou, F. T., Gauthier, P., . . . Mouahid, G. (2021). *Mastomys natalensis* (Smith, 1834) as a natural host for *Schistosoma haematobium* (Bilharz, 1852) Weinland, 1858 x *Schistosoma bovis* Sonsino, 1876 introgressive hybrids. *Parasitology Research*, 120(5), 1755-1770. doi:10.1007/s00436-021-07099-7
- Schirò, G., Pieri, D., Valvo, M. L., Gradoni, L., Cacciò, S. M., Severini, F., . . . Loria, G. R. (2021). Monitoring campaign over an edible dormouse population (*Glis glis*; Rodentia: Gliridae) in Sicily: First report of mesocestodiasis. *Animals*, 11(4) doi:10.3390/ani11040934
- Sharifdini, M., Norouzi, B., Azari-Hamidian, S., & Karamzadeh, N. (2021). The first record of ectoparasites of raccoons (*Procyon lotor*) (Carnivora, Procyonidae) in Iran. *Persian Journal of Acarology*, 10(1), 41-54. doi:10.22073/pja.v10i1.63915

- Silva, A. C. S., Paschoal, A. T. P., Bernardes, J. C., de Matos, A. M. R. N., Balbino, L. S., Santomauro, R. A., . . . Pinto-Ferreira, F. (2021). Parasites in road-killed wild felines from north of Paraná state, Brazil. [Parasitas em felinos selvagens mortos em estradas do norte do estado do paran , brasil] *Revista Brasileira De Parasitologia Veterinaria*, 30(1), 1-10. doi:10.1590/s1984-296120201090
- Silva, T. T. P., Teixeira, A. V., & Silva, A. D. A. E. (2021). Pyriproxyfen ingested with attractive toxic sugar baits is carried by the feces and reduces the reproductive potential of adult female *Aedes aegypti* (Diptera: Culicidae). *Journal of Medical Entomology*, 58(3), 1264-1269. doi:10.1093/jme/tjaa265
- Sinclair, B. J., Macleod, A. M., & Wheeler, T. A. (2021). Revision of the nearctic genus *Tritoxa* loew (Diptera: Ulidiidae). *Zootaxa*, 4920(3) doi:10.11646/zootaxa.4920.3.3
- Sinclair, P. H., Mossop, M. D., & Stotyn, S. A. (2021). Nesting ecology and reuse of nest burrows by bank swallow (*Riparia riparia*) in southern Yukon. *Canadian Field-Naturalist*, 134(4), 329-341. doi:10.22621/CFN.V134I4.2427
- Smith, J. E., Smith, I. B., Working, C. L., Russell, I. D., Krout, S. A., Singh, K. S., & Sih, A. (2021). Host traits, identity, and ecological conditions predict consistent flea abundance and prevalence on free-living California ground squirrels. *International Journal for Parasitology*, 51(7), 587-598. doi:10.1016/j.ijpara.2020.12.001
- Souza, U. A., Webster, A., Dall'Agnol, B., Peters, F. B., Favarini, M. O., Schott, D., . . . Soares, J. F. (2021). Ticks, mites, fleas, and vector-borne pathogens in free-ranging neotropical wild felids from southern Brazil. *Ticks and Tick-Borne Diseases*, 12(4) doi:10.1016/j.ttbdis.2021.101706
- Stavtseva, N. A., Fielden, L. J., Khokhlova, I. S., Warburton, E. M., van der Mescht, L., & Krasnov, B. R. (2021). Colonization of a novel host by fleas: Changes in egg production and egg size. *Parasitology Research*, 120(2), 451-459. doi:10.1007/s00436-021-07052-8
- STEKOLNIKOV, A. A. (2021). A checklist of chigger mites (Acariformes: Trombiculidae) of southeast Asia. *Zootaxa*, 4913(1), 1-163. doi:10.11646/zootaxa.4913.1.1
- Steventon, C., Koehler, A. V., Dobson, E., Wicker, L., Legione, A. R., Devlin, J. M., . . . Gasser, R. B. (2021). Detection of *Breinlia* sp. (Nematoda) in the leadbeater's possum (*Gymnobelideus leadbeateri*). *Intl. J. Parasitol.: Parasites and Wildlife*, 15, 249-254. doi:10.1016/j.ijppaw.2021.06.002
- Storari, A. P., Rodrigues, T., Bantim, R. A. M., Lima, F. J., & Saraiva, A. A. F. (2021). Mass mortality events of autochthonous faunas in a lower Cretaceous Gondwanan lagerst tte. *Scientific Reports*, 11(1) doi:10.1038/s41598-021-85953-5
- Sun, H., Ding, L., Yan, L., Pape, T., & Zhang, D. (2021). *Ascodiapteron sanmingensis* sp. nov., a new bat fly (Hippoboscidae: Streblid grade) from Fujian, China. *Biodiversity Data Journal*, 9, 1-13. doi:10.3897/BDJ.9.E64558
- Suntsov, V. V. (2021). Genomogenesis of the plague bacteria *Yersinia pestis* as a process of mosaic evolution. *Russian Journal of Genetics*, 57(2), 139-151. doi:10.1134/S1022795421020113
- Tan, L. P., Hamdan, R. H., Hassan, B. N. H., Reduan, M. F. H., Okene, I. A. -, Loong, S. K., . . . Lee, S. H. (2021). *Rhipicephalus* tick: A contextual review for southeast Asia. *Pathogens*, 10(7) doi:10.3390/pathogens10070821
- Thoroughgood, J. T., Armstrong, J. S., White, B., Anstead, C. A., Galloway, T. D., Lindsay, L. R., . . . Chilton, N. B. (2021). Molecular differentiation of four species of *Oropsylla* (Siphonaptera:

Ceratophyllidae) using PCR-based single strand conformation polymorphism analyses and DNA sequencing. *Journal of Medical Entomology*, 58(1), 241-245. doi:10.1093/jme/tjaa161

Tielemans, E., Buellet, P., Young, D., Viljoen, A., Liebenberg, J., & Prullage, J. (2021). Efficacy of a novel topical combination of Esafoxolaner, Eprinomectin and Praziquantel against adult cat flea *Ctenocephalides felis* and flea egg production in cats. *Parasite*, 28 doi:10.1051/parasite/2021017

Tielemans, E., Otsuki, T., Cheesman, T., Selmes, F., Pfefferkorn, A., & Prullage, J. (2021). Efficacy of a novel topical combination of Esafoxolaner, Eprinomectin and Praziquantel against fleas in cats, under field conditions. *Parasite*, 28 doi:10.1051/parasite/2021018

Tlapaya-Romero, L., Santos-Moreno, A., & Ibáñez-Bernal, S. (2021). Effect of seasonality and microclimate on the variation in bat-fly load (Diptera: Streblidae) in a cave bat assemblage in a dry forest. *Mammalia*, 85(4), 345-354. doi:10.1515/mammalia-2020-0115

Tran, A., Minter, G. L., Balleydier, E., Etheves, A., Laval, M., Boucher, F., . . . Tortosa, P. (2021). Describing fine spatiotemporal dynamics of rat fleas in an insular ecosystem enlightens abiotic drivers of murine typhus incidence in humans. *PLoS Neglected Tropical Diseases*, 15(2), 1-16. doi:10.1371/journal.pntd.0009029

Tripp, D. W., Emslie, A. C., Sack, D. A., & Zieschang, M. (2021). A prototype insecticide applicator and quality control monitoring for plague management on prairie dog colonies. *Wildlife Society Bulletin*, 45(1), 176-183. doi:10.1002/wsb.1165

Trnka, A., Fend'a, P., Požgayová, M., & Procházka, P. (2021). Common generalist mites do not transmit from foster parents to brood parasitic chicks. *J. Zool.*, 313(3), 195-201. doi:10.1111/jzo.12847

Tsurim, I., Wasserberg, G., Ben Natan, G., & Abramsky, Z. (2021). Systemic control of cutaneous leishmaniasis sand-fly vectors: Fipronil-treated rodent bait is effective in reducing *Phlebotomus papatasi* (Diptera: Psychodidae) female emergence rate from rodent burrows. *Journal of Medical Entomology*, 58(2), 974-978. doi:10.1093/jme/tjaa201

Tsurim, I., Wasserberg, G., Ben Natan, G., & Abramsky, Z. (2021). The potential of systemic control of sand flies using Fipronil in the novel *Leishmania major* (Kinetoplastida: Trypanosomatidae) reservoirs *Meriones tristrami* (Rodentia: Muridea) and *Meriones crassus* (Rodentia: Muridea). *Journal of Medical Entomology*, 58(2), 969-973. doi:10.1093/jme/tjaa200

Urdapilleta M., Galliari, C.A., Navarro Febre, T, & M. Lareschi. 2021. Effect of host and environment related factors on the distribution of the ectoparasites of the montane grass mouse *Akodon montensis* (Cricetidae, Sigmodontinae) in the Atlantic Forest ecoregion in northeastern Argentina. *Revista Mexicana de Biodiversidad*. En prensa.

van der Mescht, L., Matthee, S., & Matthee, C. A. (2021). New taxonomic and evolutionary insights relevant to the cat flea, *Ctenocephalides felis*: A geographic perspective. *Molecular Phylogenetics and Evolution*, 155 doi:10.1016/j.ympev.2020.106990

Veitch, J. S. M., Bowman, J., Mastromonaco, G., & Schulte-Hostedde, A. I. (2021). Corticosterone response by *Peromyscus* mice to parasites, reproductive season, and age. *General and Comparative Endocrinology*, 300 doi:10.1016/j.ygcen.2020.113640

Waindok, P., Raue, K., Grilo, M. L., Siebert, U., & Strube, C. (2021). Predators in northern Germany are reservoirs for parasites of one health concern. *Parasitol. Res.*, doi:10.1007/s00436-021-07073-3

- Wang, J. -, & Hua, B. -. (2021). Morphological phylogeny of Panorpidae (Mecoptera: Panorpoidea). *Systematic Entomology*, 46(3), 526-557. doi:10.1111/syen.12474
- Wang, Q., Cao, X., Jiang, H., & Guo, Z. (2021). Straw application and soil microbial biomass carbon change: A meta-analysis. *Clean - Soil, Air, Water*, 49(2) doi:10.1002/clen.202000386
- Warburton, E. M., Khokhlova, I. S., Palme, R., Surkova, E. N., & Krasnov, B. R. (2021). Effects of ectoparasite infestation during pregnancy on physiological stress and reproductive output in a rodent-flea system. *International Journal for Parasitology*, 51(8), 659-666. doi:10.1016/j.ijpara.2020.12.005
- Wong, Serena, Hany Elsheikha, and Michael Dryden. 2021. Flea product efficacy, pet owners' adherence and treatment failure: what's the connection? *Companion Animal* 26:8, 182-190.
- Yadav, S. K., Sarkar, S., Sarkar, S., & Siddiki, A. Z. (2021). Prevalence of endoparasites and ectoparasites of captive peafowl farm. *Advances in Animal and Veterinary Sciences*, 9(3), 442-445. doi:10.17582/journal.aavs/2021/9.3.442.445
- Yang, W. -, Hsu, M. -, Shu, P. -, Tsai, K. -, & Fang, C. -. (2021). Neglected human *Rickettsia felis* infection in Taiwan: A retrospective seroepidemiological survey of patients with suspected rickettsioses. *PLoS Neglected Tropical Diseases*, 15(4) doi:10.1371/journal.pntd.0009355
- Yeeles, P., Strain, A., Lenancker, P., & Lach, L. (2021). Low reduction of invasive ant colony productivity with an insect growth regulator. *Pest Mgt. Science*, 77(4), 1626-1632. doi:10.1002/ps.6181
- Yusoff, N., Ghani, I. A., Othman, N. W., Aizat, W. M., & Hassan, M. (2021). Toxicity and sublethal effect of Farnesyl acetate on diamondback moth, *Plutella xylostella* (L.) (Lepidoptera: Plutellidae). *Insects*, 12(2), 1-12. doi:10.3390/insects12020109
- Zhang, Y., Nie, Y., Deng, Y. -, Liu, G. -, & Fu, Y. -. (2021). The complete mitochondrial genome sequences of the cat flea *Ctenocephalides felis felis* (Siphonaptera: Pulicidae) support the hypothesis that *C. felis* isolates from China and USA were the same *C. f. felis* subspecies. *Acta Tropica*, 217 doi:10.1016/j.actatropica.2021.105880
- Zhang, Y. -, Shih, P. J. M., Wang, J. -, McNamara, M. E., Shih, C., Ren, D., & Gao, T. -. (2021). Jurassic scorpionflies (Mecoptera) with swollen first metatarsal segments suggesting sexual dimorphism. *BMC Ecology and Evolution*, 21(1) doi:10.1186/s12862-021-01771-3
- Zurita, A., Benkacimi, L., El Karkouri, K., Cutillas, C., Parola, P., & Laroche, M. (2021). New records of bacteria in different species of fleas from France and Spain. *Comparative Immunology, Microbiology and Infectious Diseases*, 76 doi:10.1016/j.cimid.2021.101648
- Zurita, A., & Cutillas, C. (2021). Combination of nuclear and mitochondrial markers as a useful tool to identify *Ctenophthalmus* species and subspecies (Siphonaptera: Ctenophthalmidae). *Organisms Diversity and Evolution*, doi:10.1007/s13127-021-00500-4

